

Deformation flows for the shifted Weil family

The transfer as an all-pass filter: resonance, escaped mass, and the endpoint direction

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Abstract

A proposal for the bulk-realization program asks whether the shifted Weil family is the deformation flow of a Wilson line: the deformation of a line's expectation obeys a differential equation fixed by the variation of its endpoints together with the non-Abelian Stokes theorem, and the shifted family already obeys a differential equation of that shape. Pursuing it turns the transfer into a recognizable object, and that object reorganizes the rest. We first write the causal generator out in full, which the program's background had defined only by naming its symbol, and show that its three factors deliver the three terms of the target exactly: the shifted form is the central form with every off-diagonal kernel and every prime atom multiplied by a single hyperbolic factor. We then show that the transfer symbol is *unimodular on the critical line*, exactly and unconditionally, by the functional equation alone. The transfer is therefore an all-pass filter, and four consequences follow. Its phase derivative is the archimedean symbol, so the Weil symbol is a group delay and the density identity of the companion investigation is explained rather than computed. Its phase jumps are resonances at the zeros, each contributing a Lorentzian of width ω and fixed mass, so the Weil form is a total resonant response. Under the Riemann hypothesis it is inner, the transfer is unitary, and the contraction defect is exactly the mass transported past the leading endpoint of the interval. And a filter that changes only phase is a phase, which is why the shift direction is abelian: truncation is multiplicative on causal kernels, the transfers and their generators lie in one commutative algebra, and the non-Abelian Stokes mechanism has nothing to act on along the shift. The contraction region in the shift-length plane is then settled completely: it is monotone in the length with supremum one or infinity and nothing between, so contraction at every length is *equivalent* to the absence of zeros at distance more than ω from the critical line. The shifted family is a graded criterion, one zero-free strip per shift; under the Riemann hypothesis the region is the whole quadrant and has no boundary; and the plane is a disproof instrument and nothing else. We then identify the margin itself. Under the Riemann hypothesis it is the constant in the *lower* sampling inequality for the Riemann zeros on the Paley–Wiener space of type $L/2$ — there is no upper one, since the zeros are not relatively separated — and it is strictly positive, because the divergence of the group delay makes the form domain embed compactly and a counting argument then forbids a null vector: the unboundedness of the Wigner–Smith delay and the positivity of the margin are one fact. The zero density meets the Nyquist density at $2\pi e^L$ with an exact sample deficit $2e^L - \frac{7}{4}$, so the sampling crossover is the explicit-formula balance. But the deficit is not what the collapse measures. Replacing the zeros by the Gram points, which have the same counting function and no arithmetic in them, leaves a margin that collapses at the same superexponential rate, to within a ratio of logarithms falling towards one: **the collapse is a density effect and the margin carries almost no arithmetic information**, so reproducing it is weak evidence for a candidate source, and a candidate is correspondingly not required to produce an arithmetic rate. What remains is the endpoint direction, where the variation is rank one at the leading end, the connection is non-abelian but flat, and the generator's own length-variation is

its boundary value. We close with the operator conditions this leaves, one of which excludes the displacement operator and another of which is satisfied structurally by an open Wilson line but not by any character — and with the observation that the Weil form’s exact symmetry group, translations and reflection in the arithmetic coordinate, is precisely the residual conformal group of a ray with one end pinned, so that conformal symmetry is aligned with the target rather than in tension with it, and positivity is invariant along a conformal orbit. Nothing here constructs a source, proves Weil positivity, or assumes the Riemann hypothesis except where a statement is explicitly labelled conditional.

1 Introduction

The bulk-realization program of this repository asks for an independently positive system whose state norm is the localized Weil quadratic form [2]. Every candidate tested so far has been asked to reproduce a *value*: a weight, a pairing, a norm. That has turned out to be a weak demand. In the one setting where the search was carried out concretely, the set of reachable states was everything analytic past the unit circle, so an exhibited weight identity carries no information unless it comes with a reason the representation could not have produced a different weight [5].

This paper takes up a proposal that asks for something else. A Wilson line, when deformed, has an expectation value obeying a differential equation fixed by the variation of its endpoints together with the non-Abelian Stokes theorem. The shifted family of the program already obeys a differential equation of exactly that shape, equation (1.7) of the background [1]:

$$\frac{\partial V_{\omega,L}}{\partial \omega} = -G_{\omega,L} V_{\omega,L}, \quad V_{0,L} = I.$$

The proposal is that these are the same equation, so that the object to search for is a *deformation* rather than an operator. The attraction is structural: a generator and an initial condition determine a flow completely, so matching a flow leaves nothing to tune. It is the right kind of demand to make.

The shape of what follows

Pursuing the proposal turns the transfer into a recognizable object, and that object reorganizes everything else. The arc of the paper is this.

First the generator is written out (Section 3). The background defined $G_{\omega,L}$ by naming its symbol and inherited the realization silently from the transfer. Written out, its three factors deliver the three terms of the target exactly, and the whole shift dependence collapses to one scalar factor: the shifted form is the central form with every off-diagonal kernel multiplied by $\cosh(\omega(x - y))$ and every prime atom by $\cosh(\omega \log n)$, the diagonal unchanged (Theorem 3.1). So the equations of the background carry the *whole* Weil form, not its archimedean part.

Then the transfer is identified (Section 4). Its symbol is *unimodular on the critical line* — exactly, unconditionally, by the functional equation and the reality of ξ alone (Proposition 4.1). The transfer changes no amplitude at any frequency, only phase. It is an **all-pass filter**, and everything after this section is a consequence of that one fact.

Four consequences, in the order they fall out.

1. **The phase derivative is the archimedean symbol.** The group delay of the filter is $2\theta'(\tau) = b(\tau^2) + w_0$ (Proposition 4.3). The companion investigation found that identity as a computation between special functions; here it is the statement that the Weil symbol is a *phase derivative*, which is also why the contact constant is not a separate ingredient and why a packet at frequency N is held for a time $\sim \log N$.

2. **The phase jumps are resonances.** Each zero contributes to $|K_\omega(i\tau) - 1|^2$ a Lorentzian of height 4 and half-width ω , of mass $4\pi\omega$, so the Weil form is the total resonant response of the input (Proposition 4.4). The zeros are the filter's resonances.
3. **Under the Riemann hypothesis the defect is a boundary flux.** The filter is then inner, the transfer is unitary, and $\langle f, D_{\omega,L}f \rangle = \int_{L/2}^\infty |(V_\omega f)|^2$ (Proposition 4.5): Weil positivity on I_L says the flow pushes mass out of the leading endpoint and never pulls it in.
4. **A filter that changes only phase is a phase, hence abelian** (Section 5). Truncation to an interval is multiplicative on causal kernels, so the transfers and their generators lie in one commutative algebra and the path-ordered solution of the flow equation collapses to an ordinary exponential. *The mechanism the proposal names has, as things stand, nothing to act on, and any non-commutativity must be supplied transverse to ω .*

The shift is then identified geometrically (Section 6): it acts on inputs by the hyperbolic rotation generated by multiplication by x , so it is a deformation and its gauge-theoretic analogue is an angle — a cusp — rather than a coupling that can be dialled.

The shift-length plane is then settled completely (Section 7). The contraction region is monotone in L with supremum *one or infinity and nothing between*, so contraction at every L is *equivalent* to the absence of zeros at distance more than ω from the critical line (Corollary 7.3). The shifted family is a **graded criterion**, one zero-free strip per shift, interpolating to the Riemann hypothesis as $\omega \downarrow 0$. Under that hypothesis the region is the whole quadrant and has no boundary at all, so there is no critical path to find; what the plane is good for is the opposite, since monotonicity makes it a **disproof instrument** on which only a certified value above one carries information. The same section locates the much smaller region on which the *generator* is positive, identifies the quantity that controls its boundary as an inverse zero-placement precision, and records why that boundary is an artifact of a sufficient condition rather than a feature of ζ .

The margin is then identified, and its collapse traced to the wrong cause (Section 8). Under the Riemann hypothesis $m_L = \lambda_{\min}(Q_{0,L})$ is the constant in the *lower* sampling inequality for the Riemann zeros on the Paley–Wiener space $PW_{L/2}$ — there is no upper one, because the zeros are not relatively separated (Proposition 8.1) — and it is strictly positive, because the divergence of the group delay makes the form domain embed compactly and a counting argument then forbids a null vector (Proposition 8.2). So the unboundedness of the Wigner–Smith delay and the positivity of the margin are one fact. The zero density meets the Nyquist density at $\tau_c = 2\pi e^L$, with an exact sample deficit $2e^L - \frac{7}{4}$, so the sampling crossover is the explicit-formula balance; and the frequency profile of the obstruction is measurably universal in τ/τ_c . But the deficit is not what the collapse measures. Replacing the zeros by the **Gram points**, which have the same counting function and no arithmetic in them at all, leaves a margin that collapses at the same superexponential rate, to within a ratio of logarithms falling from 2.10 to 1.05 across the computed range. **The collapse is a density effect, and the margin carries almost no arithmetic information** — which removes the reason to ask the mechanism question in the form Section 7 poses it, and correspondingly lowers what a candidate source has to reproduce.

What remains open is the endpoint direction (Section 9). There the variation is rank one at the leading end, with the boundary value of the evolved field as its coefficient; the connection is non-abelian but flat; the generator's own L -variation is its boundary value; and the connection fails to close on the form domain by exactly the Sobolev borderline. The prime data sits in the atoms of the connection's kernel, so the surviving object is a flat connection with punctures at the prime periods.

Finally, conditions (Section 10). Three of them, one excluding the displacement operator outright, one satisfied structurally by a bilinear in unbounded endpoint operators and by no character, and one recording what the spectrum of dimensions has to carry. Conformal invariance turns out to be aligned with the target rather than in tension with it, and L is the one conformal invariant of the configuration.

What this paper does not claim

No source is constructed. No positivity is proved. No gauge theory is matched, and none is excluded. Sections 4–7 contain statements conditional on the Riemann hypothesis; each is labelled where it occurs, and Section 7 explains structurally why no argument of this kind could be a route to a proof. Section 10’s conditions are proposals, and one of them carries an identification hypothesis the program deliberately does not make. Proposition 9.5 is a proof sketch resting on a smoothing estimate quoted from [1].

Relation to the companion investigations, and to the earlier program

The conventions, the target and the criterion are those of [2, 3]; Section 2 restates them so that this paper is self-contained apart from the background [1], which it cites by equation number for the shifted family.

Two inherited facts govern the work and are not re-derived. The accumulated exclusions of this program are statements about *channel decompositions* and about the reachable states of particular representations; none touches a gauge theory, and none constrains a flow, which is not a decomposition [4]. And the contraction whose norm is Weil positivity is saturated, with a margin collapsing superexponentially in L , so no mechanism may supply a margin uniform in L [3]. Section 7 shows the flow formulation inherits that constraint rather than escaping it, and explains the mechanism behind it.

This paper also rejoins an earlier line of the wider program. The shifted-zeta storage-depth investigation [13, 14] looked for a critical path in the shift–length plane, developed the hyperbolic identity that appears here as Proposition 6.1, and introduced the Cayley coordinate in which the contraction criterion becomes positive-realness. Section 7 settles the question that investigation was asking and explains, quantitatively, why its sufficient condition could not reach the region where anything is detectable.

Finally, the distinction between lines and loops that motivates the whole investigation is not a matter of open versus closed geometry. The test functions carry no boundary conditions; the localization is by support alone. What excluded the closed benchmark was bounded spectrum [3, 5], and Section 10 restates that in the form the deformation picture needs — as a condition on a group delay.

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2 Conventions, the target, and the shifted family

2.1 The central form

All Hilbert inner products are conjugate-linear in the first argument. For $L > 0$ let $I_L = (-L/2, L/2)$ and let E_L denote extension by zero from I_L to $\mathbb{R}$; unless another domain is stated, $f \in C_c^\infty(I_L; \mathbb{C})$ and $F = E_L f$. The test functions carry no boundary conditions: the localization is by support alone. Write

$$\widehat{F}(\tau) = \int_{\mathbb{R}} e^{-i\tau x} F(x) dx, \quad U_d F(x) = F(x - d), \quad \|F\|^2 = \frac{1}{2\pi} \int_{\mathbb{R}} |\widehat{F}(\tau)|^2 d\tau, \quad (2.1)$$

and for $r > 0$

$$n_\gamma(r) = \frac{e^{-r/2}}{1 - e^{-2r}} = \sum_{n \geq 0} e^{-a_n r}, \quad a_n = 2n + \frac{1}{2}. \quad (2.2)$$

The archimedean energy, the contact constant and the pole form are

$$K[F] = \int_0^\infty [2\|F\|^2 - 2\operatorname{Re}\langle F, U_r F \rangle] n_\gamma(r) dr = \frac{1}{2\pi} \int_{\mathbb{R}} b(\tau^2) |\widehat{F}(\tau)|^2 d\tau, \quad (2.3)$$

$$b(\tau^2) = \operatorname{Re} \psi(\tfrac{1}{4} + i\tau/2) - \psi(\tfrac{1}{4}), \quad w_0 = \psi(\tfrac{1}{4}) - \log \pi = -5.37218\dots, \quad (2.4)$$

$$P_L[f] = 2 \left| \int_{I_L} f \cosh(x/2) \right|^2 - 2 \left| \int_{I_L} f \sinh(x/2) \right|^2, \quad (2.5)$$

with $\psi = \Gamma'/\Gamma$. The split in (2.4) is bookkeeping: $b(\tau^2) + w_0 = \operatorname{Re} \psi(\tfrac{1}{4} + \frac{i\tau}{2}) - \log \pi$ is the archimedean multiplier, and $\psi(\tfrac{1}{4})$ has been moved between the Dirichlet form and the contact so that $b(0) = 0$. The target is

$$Q_L[f] = K[E_L f] + w_0 \|f\|^2 + P_L[f] - 2 \sum_{m \log p < L} (\log p) p^{-m/2} \operatorname{Re}\langle F, U_{m \log p} F \rangle, \quad (2.6)$$

the coefficient being the primitive length $\log p$ and not $m \log p$. Weil's criterion, which we use as input and do not reprove, is

$$\text{RH} \iff Q_L[f] \geq 0 \quad \text{for every } L > 0 \text{ and every } f \in C_c^\infty(I_L). \quad (2.7)$$

Two facts are used repeatedly. Q_L has the spectral form

$$Q_L[f] = \sum_{\rho} \widehat{F}(\gamma_\rho) \overline{\widehat{F}(\overline{\gamma_\rho})}, \quad \gamma_\rho = -i(\rho - \tfrac{1}{2}), \quad (2.8)$$

the sum over the nontrivial zeros in symmetric order; and the right side is very much smaller than the individual terms of (2.6), which is why assembling Q_L term by term in double precision is unusable [2, 3]. Finally, Q_L is *not* a family of functionals: it is the single Weil functional restricted to $C_c^\infty(I_L)$, since $\langle F, U_r F \rangle = 0$ automatically for $|r| \geq L$. The L -dependence is a support condition on the input and nothing else. This will matter in Section 9.

2.2 The shifted family

Let

$$\xi(s) = \tfrac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad \mathcal{F}(p) = \xi(\tfrac{1}{2} + p), \quad (2.9)$$

and for $0 < \omega \leq \frac{1}{2}$ let the causal transfer have Laplace symbol

$$K_\omega(p) = \frac{\mathcal{F}(p - \omega)}{\mathcal{F}(p + \omega)}. \quad (2.10)$$

The causal realization is taken on a line $\operatorname{Re} p = \eta > 1$, where the relevant Dirichlet series converge absolutely and the denominator does not vanish. With P_L multiplication by $\mathbf{1}_L$, M_η multiplication by $e^{-\eta x}$ and $\mathcal{M}[b]$ the whole-line Fourier multiplier by b ,

$$V_{\omega,L} = M_\eta^{-1} P_L \mathcal{M}[K_\omega(\eta + i \cdot)] P_L M_\eta, \quad V_{0,L} = I \text{ (strong limit)}, \quad (2.11)$$

equivalently $V_{\omega,L} f = (k_\omega * \tilde{f})|_{I_L}$ for the inverse Laplace kernel k_ω of K_ω , supported in $[0, \infty)$. This is (1.4) of [1].

Definition 2.1 (The generator). The causal generator symbol is

$$a_\omega(p) = \frac{\mathcal{F}'(p - \omega)}{\mathcal{F}(p - \omega)} + \frac{\mathcal{F}'(p + \omega)}{\mathcal{F}(p + \omega)}, \quad \partial_\omega K_\omega = -a_\omega K_\omega, \quad (2.12)$$

and $G_{\omega,L}$ is realized from a_ω exactly as $V_{\omega,L}$ is realized from K_ω , on the same line and with the same compression:

$$G_{\omega,L} = M_\eta^{-1} P_L \mathcal{M}[a_\omega(\eta + i \cdot)] P_L M_\eta, \quad G_{\omega,L} g = (g_\omega * \tilde{g})|_{I_L}, \quad (2.13)$$

with g_ω the inverse Laplace kernel of a_ω , supported in $[0, \infty)$. Its natural domain is $\mathcal{D}_{\log,L} = \{f \in L^2(I_L) : \int \log(2 + |\tau|) |\hat{f}|^2 < \infty\}$, on which the forms below are closed and semibounded.

The operator $G_{\omega,L}$ is not symmetric. Only its symmetric part is a form, and that form is the shifted target:

$$\operatorname{Re} \langle g, G_{\omega,L} g \rangle = Q_{\omega,L}[g], \quad \text{equivalently} \quad G_{\omega,L}^* + G_{\omega,L} \text{ represents } 2Q_{\omega,L}, \quad (2.14)$$

with $Q_{0,L} = Q_L$ of (2.6). Section 3 proves (2.14) by exhibiting the three factors.

2.3 The flow identities

For fixed $f \in C_c^\infty(I_L)$, held fixed as ω varies,

$$\frac{\partial V_{\omega,L}}{\partial \omega} = -G_{\omega,L} V_{\omega,L}, \quad V_{0,L} = I, \quad \frac{d}{d\omega} \|V_{\omega,L} f\|^2 = -2 Q_{\omega,L}[V_{\omega,L} f], \quad (2.15)$$

which is (1.7) of [1]. At $\omega = 0$ the first identity is a right derivative on these test functions and no operator-norm derivative is asserted. With the contraction defect $D_{\omega,L} = I - V_{\omega,L}^* V_{\omega,L}$,

$$\frac{\partial D_{\omega,L}}{\partial \omega} = V_{\omega,L}^* (G_{\omega,L}^* + G_{\omega,L}) V_{\omega,L}, \quad D_{0,L} = 0, \quad (2.16)$$

$$\langle f, D_{\omega,L} f \rangle = 2 \int_0^\omega Q_{s,L}[V_{s,L} f] ds, \quad Q_{0,L}[f] = \lim_{\omega \downarrow 0} \frac{\langle f, D_{\omega,L} f \rangle}{2\omega}, \quad (2.17)$$

which are (1.8) and (1.9) of [1]. Nonnegativity of $D_{\omega,L}$ is exactly $\|V_{\omega,L}\| \leq 1$. The route the deformation picture would have to serve is the following, (Proposition 1.1 of [1]):

Proposition 2.2 (A sufficient contraction route). *If there exist $L_j \rightarrow \infty$ and $0 < \omega_j \downarrow 0$ with $\|V_{\omega_j,L_j}\| \leq 1$ for every j , then the Riemann hypothesis holds.*

We stress two things about this route, because both bear on Section 11. Positivity of $Q_{\omega,L}$ at all shifts is *not* the criterion (2.7); it is a strictly stronger requirement on the family. And the cumulative defect (2.17) can be positive on an input where the instantaneous form is negative, so the two conditions are genuinely different.

3 The generator, and what it contains

Definition 2.1 names an operator; this section identifies it. The outcome is that (2.14) holds with the *complete* target on the right, and that the whole ω -dependence is one scalar factor. Section 4 then identifies the transfer, and everything after that is a consequence.

Theorem 3.1 (The shifted form in closed form). *For $0 \leq \omega \leq \frac{1}{2}$ and $f \in C_c^\infty(I_L)$, $F = E_L f$,*

$$\begin{aligned} Q_{\omega,L}[f] = & \int_0^\infty [2\|F\|^2 - 2 \cosh(\omega r) \operatorname{Re}\langle F, U_r F \rangle] n_\gamma(r) \, dr + w_0 \|f\|^2 \\ & + 2 \iint_{I_L^2} \cosh(\omega(x-y)) \cosh\left(\frac{x-y}{2}\right) \overline{f(x)} f(y) \, dx \, dy \\ & - 2 \sum_{m \log p < L} (\log p) p^{-m/2} \cosh(\omega m \log p) \operatorname{Re}\langle F, U_{m \log p} F \rangle. \end{aligned} \quad (3.1)$$

At $\omega = 0$ this is (2.6). In words: the shifted form is the central form with every off-diagonal kernel multiplied by $\cosh(\omega(x-y))$ and every prime atom by $\cosh(\omega \log n)$, the diagonal unchanged.

Proof. Write $\mathcal{F}(p) = \xi(\frac{1}{2} + p)$ and $s = \frac{1}{2} + p$. Logarithmic differentiation of (2.9) gives

$$\frac{\mathcal{F}'}{\mathcal{F}}(p) = \underbrace{\frac{1}{p + \frac{1}{2}} + \frac{1}{p - \frac{1}{2}}}_{\frac{1}{2}s(s-1)} \underbrace{-\frac{1}{2} \log \pi + \frac{1}{2} \psi\left(\frac{1}{4} + \frac{p}{2}\right)}_{\pi^{-s/2} \Gamma(s/2)} + \underbrace{\frac{\zeta'}{\zeta}\left(\frac{1}{2} + p\right)}_{\zeta(s)}, \quad (3.2)$$

and $a_\omega(p)$ of (2.12) is the sum of (3.2) at $p - \omega$ and at $p + \omega$. We compute the causal kernel of each factor on $\operatorname{Re} p = \eta > 1$ and then symmetrize. For a causal convolution with kernel c and g supported in I_L ,

$$\operatorname{Re}\langle g, (c * \tilde{g})|_{I_L} \rangle = \frac{1}{2} \iint_{x \neq y} c(|x-y|) \overline{g(x)} g(y) \, dx \, dy, \quad (3.3)$$

since relabelling $x \leftrightarrow y$ in the conjugate term turns the region $y < x$ into $x < y$ with $c(|x-y|)$ unchanged.

Pole factors. The four terms $\frac{1}{p \mp \omega \pm \frac{1}{2}}$ have simple poles at $p = \pm \frac{1}{2} \pm \omega$ — poles of the individual terms, which cancel in the sum, as Remark 4.2 records — all to the left of $\operatorname{Re} p = \eta$ since $\eta > 1 \geq \frac{1}{2} + \omega$. Because $\frac{1}{p-a}$ is the Laplace transform of $e^{ax} \mathbf{1}_{x>0}$, their kernel is

$$(e^{x/2} + e^{-x/2})(e^{\omega x} + e^{-\omega x}) \mathbf{1}_{x>0} = 4 \cosh(x/2) \cosh(\omega x) \mathbf{1}_{x>0},$$

and (3.3) turns it into $2 \iint \cosh(\omega(x-y)) \cosh(\frac{x-y}{2}) \overline{f} f$, the third line of (3.1). At $\omega = 0$ the expansion $2 \cosh \frac{x-y}{2} = 2 \cosh \frac{x}{2} \cosh \frac{y}{2} - 2 \sinh \frac{x}{2} \sinh \frac{y}{2}$ returns $P_L[f]$ of (2.5).

Zeta factor. On $\operatorname{Re} p = \eta > 1$ the Dirichlet series converges absolutely, so

$$\frac{\zeta'}{\zeta}\left(\frac{1}{2} + p - \omega\right) + \frac{\zeta'}{\zeta}\left(\frac{1}{2} + p + \omega\right) = -2 \sum_{n \geq 2} \Lambda(n) n^{-1/2} \cosh(\omega \log n) n^{-p},$$

and $n^{-p} = e^{-p \log n}$ is the Laplace symbol of the causal delay by $\log n$. Applying (3.3) to the atom at $r = \log n$ gives $-2 \Lambda(n) n^{-1/2} \cosh(\omega \log n) \operatorname{Re}\langle F, U_{\log n} F \rangle$; writing $n = p^m$, $\Lambda(p^m) = \log p$ gives the last line of (3.1). The cutoff $m \log p < L$ is automatic, because a delay of length $\geq L$ has zero action on I_L .

Gamma and π factors. Their contribution to a_ω is $-\log \pi + \frac{1}{2}[\psi(\frac{1}{4} + \frac{p-\omega}{2}) + \psi(\frac{1}{4} + \frac{p+\omega}{2})]$. With $\psi(z) = -\gamma_E + \int_0^\infty (e^{-t} - e^{-zt})(1 - e^{-t})^{-1} dt$ [8, §5.9] at $z = \frac{1}{4} + \frac{p \pm \omega}{2}$ and the substitution $t = 2r$,

$$\frac{1}{2} \left[\psi\left(\frac{1}{4} + \frac{p-\omega}{2}\right) + \psi\left(\frac{1}{4} + \frac{p+\omega}{2}\right) \right] = -\gamma_E + \int_0^\infty \frac{2e^{-2r}}{1 - e^{-2r}} dr - \int_0^\infty 2 \cosh(\omega r) n_\gamma(r) e^{-pr} dr, \quad (3.4)$$

using $n_\gamma(r) = e^{-r/2}(1 - e^{-2r})^{-1}$ of (2.2). The two integrals in (3.4) diverge logarithmically at the origin and only their difference converges, both integrands being $\sim \frac{1}{2r}$; the pairing of the divergent constant against $\|f\|^2$ with the divergent convolution against $\text{Re}\langle F, U_r F \rangle$ is exactly the combination $2\|F\|^2 - 2 \cosh(\omega r) \text{Re}\langle F, U_r F \rangle$ appearing in the first line of (3.1), which is integrable against n_γ because it is $O(r)n_\gamma(r)$ at the origin and $2\|F\|^2 n_\gamma(r)$ for $r \geq L$. This fixes the first line up to an ω -independent constant multiple of $\|f\|^2$, and that constant is $w_0 = \psi(\frac{1}{4}) - \log \pi$ by evaluating at $\omega = 0$, $p = i\tau$, where the multiplier of (2.3) and (2.4) is $\text{Re} \psi(\frac{1}{4} + \frac{i\tau}{2}) - \log \pi = b(\tau^2) + w_0$.

Each of the three contributions is ω -independent on the diagonal, since $\cosh 0 = 1$; summing them gives (3.1), and at $\omega = 0$ gives (2.6). The ω -dependent part of each line is separately convergent, because $\cosh(\omega t) - 1 = O(t^2)$ at the origin. $\square$

Remark 3.2 (The equations of the background are general). Theorem 3.1 settles a question that the background leaves implicit. The flow identities (2.15)–(2.17) are statements about the *whole* Weil form: the archimedean energy, the contact, the poles and the primes all appear, and all are deformed. A separate paragraph of the background isolates the gamma and π factors alone, K_ω^Γ , and observes that a positive tower realizes only that portion of the zero-shift generator; that is a statement about a construction, not about the scope of (2.15).

Remark 3.3 (The pole term is invisible on the imaginary axis). The causal definition on $\text{Re } p = \eta > 1$ is not a technical convenience. At $p = i\tau$,

$$\text{Re} \left[\frac{2}{p - \frac{1}{2}} + \frac{2}{p + \frac{1}{2}} \right] = \frac{2 \cdot \frac{1}{2}}{\frac{1}{4} + \tau^2} + \frac{2 \cdot (-\frac{1}{2})}{\frac{1}{4} + \tau^2} = 0$$

identically in τ . Reading a_ω as a Fourier multiplier on the imaginary axis therefore recovers the archimedean term and the prime delays and *loses the contact and pole term entirely*. The pole factors reach $Q_{\omega,L}$ only through the right half-plane realization of Definition 2.1. Appendix A verifies both halves of this remark numerically: the causal kernel $4 \cosh(x/2) \mathbf{1}_{x>0}$ reproduces P_L to machine precision, and the same symbol has vanishing real part on the imaginary axis. Remark 4.2 gives the general form of the phenomenon: the contour may be moved for K_ω and may not for a_ω .

4 The transfer is an all-pass filter

Section 3 wrote the generator out. This section identifies the transfer, and the identification is the hinge of the paper: one elementary proposition, and four consequences that occupy the rest of it.

4.1 The symbol is unimodular

Proposition 4.1 (Unimodularity). *For every real τ and every ω with $\xi(\frac{1}{2} + \omega + i\tau) \neq 0$,*

$$|K_\omega(i\tau)| = 1.$$

No hypothesis is used beyond the functional equation and the reality of ξ on the real axis.

Proof. $\xi(s) = \xi(1-s)$ gives $\xi(\frac{1}{2} - \omega + i\tau) = \xi(\frac{1}{2} + \omega - i\tau)$, and ξ has real Taylor coefficients about $s = \frac{1}{2}$, so $\xi(\bar{s}) = \overline{\xi(s)}$ and $\xi(\frac{1}{2} + \omega - i\tau) = \overline{\xi(\frac{1}{2} + \omega + i\tau)}$. Numerator and denominator of (2.10) are complex conjugates. $\square$

So Θ_ω is an *all-pass filter*: it changes no amplitude at any frequency, only phase. Writing

$$\xi(\frac{1}{2} + \omega + i\tau) = R(\tau, \omega) e^{i\Psi(\tau, \omega)}, \quad K_\omega(i\tau) = e^{-2i\Psi(\tau, \omega)}, \quad (4.1)$$

the entire content of the transfer on the critical line is the single real function Ψ . Everything below is a statement about where that phase goes.

Remark 4.2 (A trap: which contour may be moved). Unimodularity forces $\operatorname{Re} a_\omega(i\tau) = -\partial_\omega \log |K_\omega(i\tau)| = 0$, which appears to make $Q_{\omega, L}$ vanish. It does not, and the reason is worth recording because the two symbols behave differently.

K_ω has poles only where $\xi(\frac{1}{2} + p + \omega) = 0$, at $p = \rho - \frac{1}{2} - \omega$. Under the Riemann hypothesis those sit on $\operatorname{Re} p = -\omega$, so the inverse Laplace contour *may* be moved from $\operatorname{Re} p = \eta > 1$ to $\operatorname{Re} p = 0$. But a_ω has poles at $p = \rho - \frac{1}{2} \pm \omega$, hence also on $\operatorname{Re} p = +\omega$, inside the right half-plane, so for it the contour *may not* be moved. The continuation $a_\omega(i\tau)$ is therefore not the symbol of the causal operator, and the residues crossed in the shift are the whole of the form: as $\omega \downarrow 0$ they become $2\pi \sum_\rho \delta(\tau - \gamma_\rho)$, which is (2.8).

Relatedly, the four apparent poles at $p = \pm \frac{1}{2} \pm \omega$ in the factor decomposition (3.2) *cancel* between the three factors: ψ has a pole at $p = -\frac{1}{2}$ and ζ'/ζ one at $p = \frac{1}{2}$, with matching residues. The proof of Theorem 3.1 is unaffected, because it transforms each factor on $\operatorname{Re} p = \eta > 1$ where all converge and sums; but a_ω does not have those poles.

4.2 Consequence one: the Weil symbol is a group delay

Proposition 4.3 (Group delay). *The group delay of the transfer is*

$$-\frac{d}{d\tau} \arg K_\omega(i\tau) = 2 \partial_\tau \Psi = 2 \operatorname{Re} \frac{\xi'}{\xi} \left(\frac{1}{2} + \omega + i\tau \right),$$

and $\partial_\tau \Psi$ is π times the density of zeros. Its smooth part is $\theta'(\tau)$, θ the Riemann–Siegel theta function, so the group delay is $2\theta'(\tau) = b(\tau^2) + w_0$.

Proof. The first equality is (4.1); the second is $\partial_\tau \arg \xi = \operatorname{Re}(\xi'/\xi)$ along the vertical line. That $\partial_\tau \arg \xi(\frac{1}{2} + \omega + i\tau)$ is π times the zero density is the argument principle, and $N(T) = \theta(T)/\pi + 1 + S(T)$ identifies the smooth part. The final equality is the density identity of [3]. $\square$

The companion investigation obtained $b(\tau^2) + w_0 = 2\theta'(\tau)$ as an identity between special functions and drew from it that the contact constant "is not an independent object requiring a mechanism". Proposition 4.3 says why: the archimedean symbol is a *phase derivative*, and a phase has no separate ingredients. It also reads the unboundedness rule of [3] as a statement about time rather than about divergence: the delay grows like $\log \tau$, so a packet at frequency N is held for $\sim \log N$, which is $Q_L[\chi e^{iNx}]/\log N \rightarrow \|\chi\|^2$.

4.3 Consequence two: the zeros are resonances

Proposition 4.4 (Resonant line shape). *As $\omega \downarrow 0$, near each γ_ρ ,*

$$|K_\omega(i\tau) - 1|^2 = \frac{4\omega^2}{\omega^2 + (\tau - \gamma_\rho)^2} + o(1),$$

a Lorentzian of height 4 and half-width ω , of mass $4\pi\omega$. Consequently

$$\frac{1}{2\pi} \int_{\mathbb{R}} |K_{\omega}(i\tau) - 1|^2 |\widehat{F}(\tau)|^2 d\tau \longrightarrow 2\omega \sum_{\rho} |\widehat{F}(\gamma_{\rho})|^2 = 2\omega Q_{0,L}[f]. \quad (4.2)$$

Proof. Near a simple zero, $\xi(\frac{1}{2} + \omega + i\tau) \approx \xi'(\rho)(\omega + i(\tau - \gamma))$, and by Proposition 4.1 the numerator is its conjugate, so $K_{\omega}(i\tau)$ is a Blaschke factor times a constant phase. Requiring $K_{\omega} \rightarrow 1$ away from the zero as $\omega \downarrow 0$ fixes that phase, leaving $K_{\omega}(i\tau) \approx (\omega - i(\tau - \gamma))/(\omega + i(\tau - \gamma))$, for which $|K_{\omega} - 1|^2$ is the stated Lorentzian exactly; its mass is $\int 4\omega^2(\omega^2 + u^2)^{-1} du = 4\pi\omega$. Summing over the zeros and dividing by 2π gives (4.2), whose right side is (2.8). $\square$

So the leading behaviour of the defect, $\langle f, D_{\omega,L}f \rangle = 2\omega Q_{0,L}[f] + o(\omega)$, is not a formal consequence of the flow identities but a statement about line shapes: each zero is a resonance of width ω and fixed integrated strength, and the Weil form is the total resonant response of the input. Appendix A verifies the line shape, its mass and the window sum in a model built from published zero ordinates, where all three hold exactly.

4.4 Consequence three: under RH the defect is a boundary flux

Proposition 4.5 (Inner, unitary, and the flux identity). *Assume the Riemann hypothesis. Then K_{ω} is an inner function of the right half-plane, with zeros exactly at $p = \omega + i\gamma_{\rho}$; V_{ω} is unitary on $L^2(\mathbb{R})$ and causal; and for f supported in I_L ,*

$$\langle f, D_{\omega,L}f \rangle = \|(1 - P_L)V_{\omega}f\|^2 = \int_{L/2}^{\infty} |(V_{\omega}f)(x)|^2 dx. \quad (4.3)$$

Proof. Under RH the poles of K_{ω} lie on $\text{Re } p = -\omega$, so K_{ω} is analytic on $\text{Re } p \geq 0$; it is bounded there, the Γ -ratio giving $K_{\omega}(p) = O(|p|^{-\omega})$ while the ζ and pole ratios tend to 1; and it is unimodular on the boundary by Proposition 4.1. Hence it is inner, the Blaschke condition $\sum_{\rho} \omega/(1 + \gamma_{\rho}^2) < \infty$ holding. No pole lies between $\text{Re } p = \eta$ and $\text{Re } p = 0$, so the inverse Laplace contour may be moved to the imaginary axis and V_{ω} is the Fourier multiplier by $K_{\omega}(i\cdot)$, of modulus one, hence unitary. Then $\|f\|^2 - \|P_L V_{\omega}f\|^2 = \|V_{\omega}f\|^2 - \|P_L V_{\omega}f\|^2 = \|(1 - P_L)V_{\omega}f\|^2$, and causality gives $\text{supp } V_{\omega}f \subseteq [-L/2, \infty)$, so all the escaping mass is to the right. $\square$

(4.3) is the sharpest statement in the paper. The contraction defect is not an abstract operator quantity: it is the **flux of $|V_{\omega}f|^2$ through the leading endpoint of the interval**, and Weil positivity on I_L says the flow pushes mass out of the moving end and never pulls it in. Section 9 finds the same point from the geometry, as the location of a rank-one variation.

Remark 4.6 (Direction of the hypothesis). Proposition 4.5 uses RH; it is the easy direction of the program made explicit, not a route to a proof. Its value is that it exhibits the mechanism, which Sections 7 and 9 then use.

4.5 The scattering dictionary

Collected, Propositions 4.1–4.5 are a definition.

Established here	Scattering name
k_ω supported in $[0, \infty)$; K_ω analytic in a right half-plane	causality, via Paley–Wiener
$ K_\omega(i\tau) = 1$, exactly and unconditionally	unitarity on the energy shell
poles at $-\omega + i\gamma_\rho$ (Remark 4.2)	resonances of width ω
$-d\varphi/d\tau = 2\theta'(\tau)$ (Prop. 4.3)	Wigner–Smith time delay = density of states [25]
$\Psi/\pi \rightarrow N(\tau)$ (Prop. 4.3)	Krein spectral shift function [24]

Two further points make the identification specific. K_ω is a *ratio* — ξ at the reflected argument over ξ at the argument, with the functional equation supplying the reflection — and a unimodular ratio of that shape is what a one-channel *reflection coefficient* is; a half-infinite line with a boundary condition at one end has exactly one scattering channel and its entire S -matrix is such a coefficient. And the argument-principle bookkeeping of Proposition 4.3 is Levinson’s theorem, the total phase change per resonance being -2π .

Remark 4.7 (Scope of this reading). Reading τ as an energy and x as a scattering coordinate is an identification the program does not make; the caveat attached to the channel-tower rule of [3] applies here in full, and Section 10 gives the first motivated reason to make it without removing it. The neighbourhood is also crowded: zeros-as-resonances is old. Nothing here claims the idea. What is this program’s own is that the object arrives from its own shifted transfer, with the constants fixed — the delay is $2\theta'$ exactly and the resonance widths are exactly ω .

5 The shift direction is abelian

The proposal’s mechanism is the non-Abelian Stokes theorem: path ordering along a deformed line may be traded for surface ordering of the parallel-transported curvature, which is what converts a deformation into a differential equation with a local, gauge-invariant right-hand side. This section shows that the mechanism has nothing to act on along ω .

Proposition 4.1 says why in one word. An all-pass filter changes only phase, and a phase is a scalar; a one-parameter family of them can carry no ordering. The proof below is independent of that section and uses only causality, which is worth having because it locates the property in the truncation rather than in the arithmetic — but the reader who wants the reason rather than the proof should take it from Section 4.

Lemma 5.1 (Truncation is multiplicative on causal kernels). *Let A, B be convolution operators whose kernels are supported in $[0, \infty)$, and let P_L be multiplication by $\mathbf{1}_{I_L}$. Then*

$$P_L A P_L B P_L = P_L A B P_L.$$

Proof. Let f be supported in I_L and put $g = Bf$. Causality of B gives $\text{supp } g \subseteq [-L/2, \infty)$. For $x \in I_L$, causality of A gives $(Ag)(x) = \int_0^\infty a(t)g(x-t)dt$, which involves g only on $(-\infty, x]$, hence only on $[-L/2, x] \subseteq I_L$. Replacing g by $P_L g$ therefore does not change $(Ag)(x)$ for $x \in I_L$, which is the claim. $\square$

Theorem 5.2 (The flow is abelian). *Fix $L > 0$. The operators $\{V_{\omega,L}\}_{0 \leq \omega \leq 1/2}$ and $\{G_{\omega,L}\}_{0 \leq \omega \leq 1/2}$ all lie in one commutative algebra. In particular*

$$[V_{\omega,L}, V_{\omega',L}] = [G_{\omega,L}, G_{\omega',L}] = [G_{\omega,L}, V_{\omega',L}] = 0 \quad \text{for all } \omega, \omega',$$

and the solution of the flow equation (2.15) carries no path ordering:

$$V_{\omega,L} = \exp \left(- \int_0^\omega G_{s,L} ds \right), \quad \text{equivalently} \quad \log K_\omega = - \int_0^\omega a_s ds. \quad (5.1)$$

Proof. By (2.11) and (2.13) every one of these operators has the form $M_\eta^{-1} P_L \mathcal{M}[b(\eta + i \cdot)] P_L M_\eta$ for a symbol b analytic and bounded on $\text{Re } p = \eta$, and the conjugation by M_η replaces a kernel $k(x - y)$ by $k(x - y)e^{\eta(x-y)}$, again a causal function of $x - y$. So each is the truncation to I_L of a convolution with kernel supported in $[0, \infty)$. By Lemma 5.1 the truncation map is multiplicative on this class, hence an algebra homomorphism onto its image, and convolution is commutative. This gives the three vanishing commutators.

A commuting family of generators makes the path-ordered solution of $\partial_\omega V = -GV$, $V_0 = I$, the ordinary exponential of $-\int_0^\omega G_{s,L} ds$; equivalently, applying the homomorphism to the whole-line identity $\partial_\omega K_\omega = -a_\omega K_\omega$, $K_0 = 1$, which integrates to the scalar statement on the right of (5.1). $\square$

Remark 5.3 (Discrete form). In a discretization ordered by x , Lemma 5.1 is the remark that a truncated causal convolution is a lower-triangular Toeplitz matrix, hence a power series in one nilpotent shift, and that any two such commute. Appendix A records this as an exact rational computation, together with the control showing that causality is what does the work: truncated *non-causal* convolutions do not commute.

Corollary 5.4 (No non-Abelian Stokes content along the shift). *The shift flow is the holonomy of an abelian connection on a one-dimensional parameter interval. Its holonomy depends on the endpoints of that interval and on nothing else; there is no surface for a curvature to be integrated over, and the non-Abelian Stokes theorem degenerates to the fundamental theorem of calculus, which is the right-hand identity in (5.1).*

Remark 5.5 (What this costs the proposal, and what it asks for). Corollary 5.4 is not a refutation. It is a statement of what is missing, and it is precise: *any non-commutativity must be supplied transverse to ω* . There are exactly three places it could come from within the objects this program already has.

1. **The endpoint direction.** P_L itself is not in the algebra of Theorem 5.2; the nested projections commute with one another but not with the transfers. This is Section 9, and it is the direction the proposal itself names.
2. **The bilinear energy balance.** The second identity of (2.15) is quadratic in $V_{\omega,L}$ where the first is linear: $\partial_\omega \|Vf\|^2 = -\langle Vf, (G^* + G)Vf \rangle$ cuts the line at the deformation point, inserts an object and pairs the two halves. That is the shape of the loop equation [10], whose right-hand side is likewise bilinear in the two segments meeting at the deformation point, and in that correspondence the Weil form occupies the slot the parallel-transported curvature occupies. The caution is that the loop equation is an equation for an expectation and is nonlinear in the loop functional, whereas (2.15) is linear at the operator level; a matching attempt must say which of the two it claims.
3. **An internal index.** The prime coefficients of (3.1) are scalars. The corresponding sum for an Artin or automorphic L -function carries $\text{tr } \rho(\text{Frob}_p)$, a trace of a holonomy in a representation, which is the one place in this subject where a non-abelian holonomy is not an analogy. We record this and do not pursue it: it changes the target from ζ to a family and nothing in the investigation so far requires it.

Remark 5.6 (Relation to the two-dimensional control). An earlier calculation in this program used rigorously constructed two-dimensional Yang–Mills [9] to reject literal Wilson-loop winding as the prime repetition law, on the ground that winding is not transfer evolution [6]. Theorem 5.2 is a statement about the transfer evolution itself and is independent of that calculation; the two are complementary, the first saying what the evolution is not and the second saying what structure the evolution does not have.

6 The shift is a hyperbolic rotation

Theorem 3.1 has an immediate consequence that identifies the shift geometrically. Let X be multiplication by x , so that $\cosh(\omega X)$ and $\sinh(\omega X)$ map $C_c^\infty(I_L)$ to itself.

Proposition 6.1 (Hyperbolic form of the shift). *For $0 \leq \omega \leq \frac{1}{2}$ and $f \in C_c^\infty(I_L)$,*

$$Q_{\omega,L}[f] = Q_{0,L}[\cosh(\omega X)f] - Q_{0,L}[\sinh(\omega X)f]. \quad (6.1)$$

Proof. Write $g_c = \cosh(\omega X)f$, $g_s = \sinh(\omega X)f$, $G_c = E_L g_c$, $G_s = E_L g_s$, and evaluate (2.6) on each, term by term. For the norms, $|\cosh(\omega x)|^2 - |\sinh(\omega x)|^2 = 1$ pointwise, so $\|g_c\|^2 - \|g_s\|^2 = \|f\|^2$; this disposes of the contact term and of the $2\|F\|^2$ inside (2.3). For a translation, at any r ,

$$\begin{aligned} \operatorname{Re}\langle G_c, U_r G_c \rangle - \operatorname{Re}\langle G_s, U_r G_s \rangle &= \operatorname{Re} \int [\cosh(\omega x) \cosh(\omega(x-r)) \\ &\quad - \sinh(\omega x) \sinh(\omega(x-r))] \overline{F(x)} F(x-r) dx \\ &= \cosh(\omega r) \operatorname{Re}\langle F, U_r F \rangle, \end{aligned}$$

by the addition formula $\cosh a \cosh b - \sinh a \sinh b = \cosh(a-b)$. Applying this at r inside the n_γ integral and at $r = m \log p$ in the prime sum reproduces the first and last lines of (3.1). For the pole form, $P_L[g] = 2 \iint \cosh(\frac{x-y}{2}) \overline{g(x)} g(y)$, and the same addition formula applied to $\cosh(\omega x) \cosh(\omega y) - \sinh(\omega x) \sinh(\omega y) = \cosh(\omega(x-y))$ reproduces the third line. Each difference is computed inside a convergent integral, so no regularization is needed. Summing gives (3.1), which is $Q_{\omega,L}[f]$ by Theorem 3.1. $\square$

Remark 6.2 (Attribution). Equation (6.1) is (1.10) of [1], and it was obtained independently, with the coordinate centred at the interval's midpoint and with the congruence $Q_{\omega,L} = C_\omega(Q - T_\omega Q T_\omega) C_\omega$ and the exact criterion $\|\tanh(\omega X)\|_{Q \rightarrow Q} \leq 1$, in the shifted-zeta storage-depth investigation [13], which has priority. The proof above derives it from the generator rather than from a kernel expansion; the two agree. That investigation's $\kappa_L = \|X\|_{Q \rightarrow Q}$ is analysed in Section 7.

The derivation from the generator is what makes the next remark available.

Remark 6.3 (The shift is an angle, not a coupling). The pair $(\cosh(\omega X), \sinh(\omega X))$ is a hyperbolic rotation by angle ω with generator X . Two consequences are worth stating.

First, the shift is a *geometric* deformation of the input, not a change of coefficients, so it is the kind of parameter a line can carry. In gauge theory the natural object with a hyperbolic angle attached to a line is a *cuspid*; this is where we would look first among specific theories, and the differential equation a cusped line's expectation satisfies in its angle has exactly the shape the proposal asks for [11].

Second, the parameter is not free. A cusp angle is a geometric datum of the contour, so a mechanism built on it cannot be dialled to fit; that is the property the program wants, since any mechanism with a tunable margin uniform in L proves something false [3]. The obstruction to

state plainly is one of type: a cusp anomalous dimension is a number, while $Q_{\omega,L}$ is a form on an infinite-dimensional input space, so a line carrying it must carry an infinite-dimensional label. This is Condition 10.1 below.

Remark 6.4 (Positivity at positive shift is a different statement). Proposition 6.1 exhibits $Q_{\omega,L}$ as a *difference* of two central energies. Nonnegativity of each does not give nonnegativity of the difference, so positivity of the shifted form at all shifts is strictly stronger than the criterion (2.7) and is not implied by the Riemann hypothesis. The route of Proposition 2.2 does not require it: what it requires is $\|V_{\omega_j,L_j}\| \leq 1$, a statement about the cumulative defect (2.17) rather than about its integrand.

7 The contraction region in the shift–length plane

The route this program must serve is Proposition 2.2: contraction along a sequence with $L_j \rightarrow \infty$ and $\omega_j \downarrow 0$. It is natural to ask what the region

$$\mathcal{R} = \{(\omega, L) : \|V_{\omega,L}\| \leq 1\}$$

looks like, and the earlier shifted-zeta investigation [13, 14] looked for a *critical path* inside it. Proposition 4.1 settles the question completely, and the answer is that the region has no shape worth computing — which is itself informative, and which converts the plane into an instrument of a different kind.

7.1 The region is monotone, and its supremum is one or infinity

Proposition 7.1 (Monotonicity). *For each fixed ω , $L \mapsto \|V_{\omega,L}\|$ is nondecreasing, and $\sup_L \|V_{\omega,L}\| = \|V_{\omega}\|_{L^2(\mathbb{R})}$.*

Proof. $V_{\omega,L} = P_L V_{\omega} P_L$ and the intervals are nested, so for $L < L'$ the first is a compression of the second. $P_L \rightarrow I$ strongly, so $\|P_L V_{\omega} P_L f\| \rightarrow \|V_{\omega} f\|$ for each f . $\square$

Proposition 7.2 (Dichotomy). *Fix $\omega \in (0, \frac{1}{2}]$ and suppose no zero of ξ has $|\operatorname{Re} \rho - \frac{1}{2}| = \omega$ exactly. Then*

$$\sup_L \|V_{\omega,L}\| = \begin{cases} 1, & \text{if } \xi \text{ has no zero with } |\operatorname{Re} \rho - \frac{1}{2}| > \omega, \\ \infty, & \text{otherwise.} \end{cases}$$

Proof. The poles of K_{ω} are at $p = \rho - \frac{1}{2} - \omega$, so one lies in $\operatorname{Re} p > 0$ exactly when some $\operatorname{Re} \rho > \frac{1}{2} + \omega$; by the functional equation that is the same as some $|\operatorname{Re} \rho - \frac{1}{2}| > \omega$. If there is none, the argument of Proposition 4.5 applies verbatim with RH replaced by that weaker hypothesis: K_{ω} is inner and V_{ω} is unitary, of norm exactly one. If there is one, at $\operatorname{Re} p = \delta - \omega > 0$, moving the contour picks up a residue and k_{ω} acquires a component growing like $e^{(\delta-\omega)x}$, so V_{ω} is unbounded on $L^2(\mathbb{R})$; by Proposition 7.1 the truncations are then unbounded in L . $\square$

Corollary 7.3 (The shifted family is a graded criterion). *For each $\omega \in (0, \frac{1}{2}]$ the following are equivalent:*

1. $\|V_{\omega,L}\| \leq 1$ for every $L > 0$;
2. ξ has no zero with $|\operatorname{Re} \rho - \frac{1}{2}| > \omega$.

In particular, under the Riemann hypothesis $\mathcal{R}$ is the whole quadrant $(0, \frac{1}{2}] \times (0, \infty)$.

So the family is not one criterion but a totally ordered family of them, one per shift, interpolating between the trivial statement at $\omega = \frac{1}{2}$ — no zeros outside the critical strip — and the Riemann hypothesis as $\omega \downarrow 0$. **Each ω buys exactly a zero-free strip of width ω .**

Remark 7.4 (The two limits of Proposition 2.2 do different jobs). $L_j \rightarrow \infty$ realizes the supremum of Proposition 7.1 and covers every test function; $\omega_j \downarrow 0$ closes the detection threshold of Corollary 7.3. Neither substitutes for the other.

7.2 There is no critical path, and what the plane is for

Corollary 7.3 says $\mathcal{R}$ has no interesting boundary. Under the Riemann hypothesis it is everything; if the hypothesis fails at distance δ , it is everything for $\omega > \delta$ and cut off at a finite L for $\omega < \delta$. The L -direction is monotone bookkeeping and the entire content sits at $L = \infty$, where it is the H^∞ norm of K_ω . There is no curve in the plane whose shape is worth computing.

This also answers the question the manuscript previously left open, whether the flow formulation escapes the saturation constraint of [3]. It does not, and the reason is at the level of quantifiers: a mechanism producing a strict contraction at one fixed $\omega > 0$, however uniform in L , would establish a zero-free strip of width ω — a real and currently open theorem, and a strong one — but not the Riemann hypothesis. The quantifier missing from such a statement is $\omega \downarrow 0$, and by (4.2) the margin vanishes with ω by construction.

Monotonicity does buy one thing, and it is worth stating on its own.

Remark 7.5 (A disproof instrument). A finite computation exhibiting $\|V_{\omega,L}\| > 1$ at a single (ω, L) proves that ξ has a zero at distance more than ω from the critical line. A computation showing $\|V_{\omega,L}\| \leq 1$ at any finite L proves nothing whatever. Numerics on the shift-length plane are therefore one-sided in the same way the companion investigation's disproof test is, and should be organized as such: certified *lower* bounds on Rayleigh quotients, scanning ω downward and L upward.

7.3 The generator region, and what controls its boundary

The generator condition $Q_{\omega,L} \succeq 0$ is strictly stronger than contraction — [13] exhibits a certified input at $L = \log 7$, $\omega = 10^{-11}$ on which the generator is negative while the cumulative defect is positive — and it *does* have a boundary. That boundary is controlled by $\kappa_L = \|X\|_{Q \rightarrow Q}$, through the sufficient path condition $\omega \kappa_L < \pi/4$ of [13]. Proposition 6.1 and the spectral form identify what κ_L measures.

Proposition 7.6 (κ_L is an inverse zero-placement precision). *With $X_a f = (x - a)f$ one has $\widehat{X_a} f = i\widehat{F}' - a\widehat{F}$, so by (2.8)*

$$\kappa_L^2 = \sup_f \frac{\sum_\rho |i\widehat{F}'(\gamma_\rho) - a\widehat{F}(\gamma_\rho)|^2}{\sum_\rho |\widehat{F}(\gamma_\rho)|^2} \xrightarrow{\text{near-null } f} \sup_f \frac{\sum_\rho |\widehat{F}'(\gamma_\rho)|^2}{\sum_\rho |\widehat{F}(\gamma_\rho)|^2},$$

the ratio of derivative samples to value samples of $\widehat{F}$ at the Riemann zeros. If e_L is the ground vector of $Q_{0,L}$ and ε_L the root-mean-square distance from the zeros of $\widehat{e}_L$ to the nearest γ_ρ , then $m_L \approx \varepsilon_L^2 Q[Xe_L]$ and $\kappa_L \approx \varepsilon_L^{-1}$, so $\kappa_L \sqrt{m_L} \approx \sqrt{Q[Xe_L]}$.

The identity for κ_L^2 is exact; the relation $\kappa_L \sqrt{m_L} \approx \sqrt{Q[Xe_L]}$ is a single-parameter heuristic. It is tested by the values recorded in [13] across twenty orders of magnitude in m_L :

horizon	m_L	κ_L	$\kappa_L \sqrt{m_L}$	$\pi/(4\kappa_L)$	recorded root
log 3	5.537×10^{-8}	6.112×10^1	0.01438	1.29×10^{-2}	1.64×10^{-2}
log 5	9.293×10^{-18}	2.534×10^6	0.00773	3.10×10^{-7}	3.95×10^{-7}
log 7	6.802×10^{-28}	2.326×10^{11}	0.00607	3.38×10^{-12}	4.30×10^{-12}

$\kappa_L \sqrt{m_L}$ moves by a factor of 2.4 while m_L moves by 10^{20} , and the recorded diagnostic root sits within a factor of 1.3 of $\pi/(4\kappa_L)$ throughout. So the generator boundary is $\omega_*(L) \asymp \sqrt{m_L}$.

Remark 7.7 (The generator boundary is an artifact). $m_L = \lambda_{\min}(Q_{0,L})$ is positive precisely when Weil positivity holds on I_L , so its superexponential collapse happens *in the world the program is trying to prove*. The generator region's boundary therefore collapses under the Riemann hypothesis as well, and carries no arithmetic information; it is a property of a sufficient condition.

How far is it from anything useful? By Proposition 7.2, a zero at distance δ becomes visible on I_L only once $(\delta - \omega)L \gtrsim 1$, so the working scale is $\omega \sim 1/L$, a power law. Against $\pi/(4\kappa_L)$ the ratios are 7.1×10^1 , 2.0×10^6 and 1.5×10^{11} at the three horizons above — eleven orders of magnitude at $L = \log 7$, widening superexponentially. The sufficient condition and the needed one are not close and are diverging, which is the quantitative reason the earlier investigation's elementary extension lemma asked for extension lengths of order $e^{-\text{const}/m_L}$.

7.4 The Cayley coordinate is the positive-real criterion

[14] introduced $Z_{\omega,L} = (I - V)(I + V)^{-1}$ and $P = \frac{2}{\omega} \text{Re } Z$, with $D_{\omega,L} = \frac{\omega}{2}(I + V^*)P(I + V)$, so that $P \succeq 0 \iff \|V\| \leq 1$, and the corresponding causal symbol $a_{\text{cum},\omega} = \frac{2}{\omega}(1 - K_\omega)/(1 + K_\omega)$. Proposition 4.1 identifies it.

Proposition 7.8 (Reactance). *On the critical line,*

$$\frac{1 - K_\omega(i\tau)}{1 + K_\omega(i\tau)} = i \tan \Psi(\tau, \omega),$$

purely imaginary. Consequently $\|V_{\omega,L}\| \leq 1$ for all L is equivalent to Z_ω being accretive, that is to $a_{\text{cum},\omega}$ being a positive-real function on the right half-plane, which by Corollary 7.3 is equivalent to K_ω being inner.

Proof. Substitute $K_\omega(i\tau) = e^{-2i\Psi}$ from (4.1); the Cayley transform carries the unit circle to the imaginary axis. The equivalences are Corollary 7.3 together with the congruence of [14]. $\square$

That is the classical passivity criterion, and positive-real functions are Nevanlinna functions after rotation, with a Herglotz representation whose measure is supported on the resonances — which is Proposition 4.4 seen from the other side. Two cautions. First, Remark 4.2 applies again: $a_{\text{cum},\omega}$ is purely imaginary on the imaginary axis, so any expansion carried out there loses the entire form, and the ω^2 expansion of [14] is safe only because it is taken on a far-right Laplace line. Second, the natural tool for such a function is its Herglotz representation, not a Taylor expansion in the shift. Both point at the same literature as the companion investigation's top open item [16].

7.5 What the margin is

Combining (4.2) with the spectral form gives the margin an exact reading:

$$\lambda_{\min}(Q_{0,L}) = \min_{\|f\|=1, \text{supp } f \subseteq I_L} \sum_{\rho} |\hat{F}(\gamma_\rho)|^2,$$

that is, **how well a function supported in an interval can avoid the Riemann zeros in frequency**. Two consequences.

The band edge is explained rather than observed. The cheapest way to avoid every γ_ρ is to live below the lowest one, so near-null directions are bandlimited below $\gamma_1 = 14.1347$ — which [5] recorded as a measurement and [3] carries as the second half of its seventh selection rule. Here it is a consequence.

But bandlimiting is only the first factor. The recorded margins give $d \log m_L / dL \approx -44$ and -69 on the two available intervals, far steeper than the bandwidth- γ_1 prolate rate $-\gamma_1$ [26]. By Proposition 7.6 the second factor is the zero-placement precision: $m_L \approx \varepsilon_L^2 Q[Xe_L]$.

What kind of object m_L is, and what its collapse is caused by, is Section 8. The short answer, stated here because it changes how this subsection should be read: m_L is the lower sampling constant of the zeros on a Paley–Wiener space, it is strictly positive (Proposition 8.2), and its collapse is produced by the *density* of the zeros rather than by their arithmetic (§8.6). The zero-placement precision is therefore a correct description of the mechanism and a misleading place to look for one.

8 The margin is a sampling constant, and the density sets it

Section 7 left the margin $m_L = \lambda_{\min}(Q_{0,L})$ with an exact reading — how well a function supported in an interval can avoid the Riemann zeros in frequency — and left its collapse as the open mechanism question. This section identifies the object, proves that it is positive, says what the classical theory of such objects reaches, and then reports a measurement which changes what the mechanism question is.

8.1 The object

For $f \in C_c^\infty(I_L)$ the transform $\widehat{F}$ is entire of exponential type $L/2$ and lies in $L^2(\mathbb{R})$, that is $\widehat{F} \in PW_{L/2}$, the Paley–Wiener space; and $\|f\|^2 = \frac{1}{2\pi} \|\widehat{F}\|^2$. Under the Riemann hypothesis the spectral form (2.8) is a sum of squares, so

$$m_L = \lambda_{\min}(Q_{0,L}) = \inf \left\{ \frac{2\pi \sum_\rho |\Phi(\gamma_\rho)|^2}{\|\Phi\|^2} : \Phi \in PW_{L/2}, \Phi \neq 0 \right\} : \quad (8.1)$$

the margin is the constant in the lower sampling inequality for the Riemann zeros on $PW_{L/2}$. Weil positivity says the zeros are a set of uniqueness for $PW_{L/2}$; a positive m_L says they are a set of sampling, with a bound. The generating function of the sampling set is $\Xi(\tau) = \xi(\frac{1}{2} + i\tau)$, a Hermite–Biehler function under the hypothesis, which is why de Branges theory keeps being named in this program [16]. Section 8.4 says how much that buys.

One half of the standard definition fails, and it is worth saying so before the language is used further.

Proposition 8.1 (No upper sampling constant). *Unconditionally, $\sup\{Q_{0,L}[f] : \|f\| = 1\} = +\infty$. Equivalently the zeros are not relatively separated: $\#\{\rho : |\gamma_\rho - T| \leq \frac{1}{2}\} \rightarrow \infty$.*

Proof. By (2.6) and (2.4), $Q_{0,L}$ differs from $\frac{1}{2\pi} \int |\widehat{F}|^2(b(\tau^2) + w_0)$ by the rank-two form P_L and by the finite prime comb, both bounded; and $b(\tau^2) + w_0 = \log \frac{|\tau|}{2\pi} + O(\tau^{-2})$ is unbounded. For the second statement, the number of zeros in a unit window at height T is $\frac{1}{2\pi} \log \frac{T}{2\pi} + o(\log T)$ by Riemann–von Mangoldt. $\square$

So $\{\gamma_\rho\}$ is *not* a sampling set in the usual two-sided sense and m_L is *not* a frame bound; it is the lower constant alone. This is the same fact as the unboundedness of the group delay in Proposition 4.3, and the same fact as the channel-tower rule of [3]. The lower inequality is the half that needs no separation, for the trivial reason that $\sum_{\lambda \in \Lambda} |\Phi(\lambda)|^2$ increases when Λ does; that monotonicity is used again in §8.6.

8.2 The infimum is positive

Proposition 8.2 (Strict positivity of the sampling constant). *Assume the Riemann hypothesis. Then $m_L > 0$ for every $L > 0$, and the infimum in (8.1) is attained.*

Proof. Write $Q_{0,L} = \mathcal{A}_L + \mathcal{B}_L$, where $\mathcal{A}_L[f] = \frac{1}{2\pi} \int |\widehat{F}|^2(b(\tau^2) + w_0)$ and $\mathcal{B}_L = P_L - (\text{prime comb})$ is bounded, of norm at most $2\|\cosh(x/2)\|_{L^2(I_L)}^2 + 2\sum_{n < e^L} \Lambda(n)n^{-1/2}$. Since $b \geq 0$ and $w_0 = -5.37218\dots$, $\mathcal{A}_L$ is bounded below, with form domain $\mathcal{D}_{\log,L}$ of Definition 2.1.

The embedding $\mathcal{D}_{\log,L} \hookrightarrow L^2(I_L)$ is compact. Let (f_k) be bounded in the form norm. Then $\widehat{F}_k$ is bounded in $L^2(\mathbb{R})$; its tails are uniformly small, $\int_{|\tau| > T} |\widehat{F}_k|^2 \leq C/\log T$; it is uniformly bounded, $|\widehat{F}_k| \leq \|f_k\|_{L^1} \leq \sqrt{L}\|f_k\|$; and it is equicontinuous, since $|\widehat{F}_k'| = |x\widehat{F}_k| \leq \frac{L^{3/2}}{2}\|f_k\|$. Arzelà–Ascoli and a diagonal argument give a subsequence converging locally uniformly, and the uniform tail bound upgrades that to convergence in $L^2(\mathbb{R})$.

Hence $\mathcal{A}_L$ has compact resolvent; adding the bounded $\mathcal{B}_L$ leaves the essential spectrum empty, so $Q_{0,L}$ has purely discrete spectrum accumulating only at $+\infty$ and its infimum is attained at some e_L with $\|e_L\| = 1$. Under the hypothesis $Q_{0,L} \geq 0$ on $C_c^\infty(I_L)$, hence on the form closure. Suppose $m_L = 0$. Along an approximating sequence $\widehat{F}_k \rightarrow \widehat{E}_L$ locally uniformly, so for every T

$$\sum_{|\gamma_\rho| < T} |\widehat{E}_L(\gamma_\rho)|^2 = \lim_k \sum_{|\gamma_\rho| < T} |\widehat{F}_k(\gamma_\rho)|^2 \leq \liminf_k Q_{0,L}[f_k] = 0,$$

so $\widehat{E}_L$ vanishes at every zero. But $\widehat{E}_L \in PW_{L/2}$ is a nonzero function of Cartwright class and type $L/2$, so the number of its zeros in $[-T, T]$ is at most $\frac{L}{\pi}T(1 + o(1))$, while the zeros of ξ supply $2N(T) = \frac{T}{\pi} \log \frac{T}{2\pi e} + \frac{7}{4} + 2S(T)$. The two are incompatible for large T . $\square$

Two remarks. The counting contradiction is the reason strict positivity holds and it is a statement about *densities*, not an estimate; attainment of the minimum is Bombieri’s [17], by a different argument, and the continuity of m_L in L is Suzuki’s [7]. What the proof above adds is the mechanism: the divergence of the group delay (Proposition 4.3) is exactly what makes the form domain embed compactly, so **the unboundedness of the Wigner–Smith delay and the positivity of the sampling constant are one fact**. An all-pass filter with bounded delay would have neither.

8.3 The sampling budget, and where it crosses

The local density of $\{\pm\gamma_\rho\}$ is $\frac{1}{2\pi} \log \frac{\tau}{2\pi}$ and the Nyquist density for $PW_{L/2}$ is $\frac{L}{2\pi}$. They meet at

$$\tau_c = 2\pi e^L, \tag{8.2}$$

below which the zeros undersample and above which they oversample; the cumulative counts cross at $T_* = e\tau_c = 2\pi e^{L+1}$, which is where the proof of Proposition 8.2 bites. The maximal shortfall is exact:

$$D(L) = \frac{L\tau_c}{\pi} - 2N(\tau_c) = \frac{\tau_c}{\pi} \left(L - \log \frac{\tau_c}{2\pi e} \right) - \frac{7}{4} = 2e^L - \frac{7}{4}, \tag{8.3}$$

the $\frac{7}{4}$ being twice the Riemann–von Mangoldt constant and S dropped. Since $e^L = X$ is the prime cutoff in (2.6), **the sampling crossover is the explicit-formula balance**: primes up to X against zeros up to height $2\pi X$, with a shortfall of $2X$ samples. This is the first place in the program where the e^L scale appears for a structural reason rather than as the range of a sum.

8.4 What classical sampling theory reaches

Very little, and it is useful to say exactly how little, because the language of the subject is being used for an object outside its hypotheses.

Landau’s necessary density condition is satisfied with room to spare, the lower uniform density being infinite. Beurling’s gap theorem is the only tool here that produces a *constant*; in the explicit form of [19] it requires the maximal gap of the sampling set to be below π/σ for type σ . The maximal gap of $\{\pm\gamma_\rho\}$ is the one straddling the origin, $2\gamma_1 = 28.2695$, and $\sigma = L/2$, so the hypothesis reads

$$L < \pi/\gamma_1 = 0.2222606\dots, \quad (8.4)$$

with constant $(\cos(\gamma_1 L/2))^{-1}$: 1.32 at $L = 0.1$ and 63 at $L = 0.22$. **Above $L \approx 0.22$ the classical machinery does not degrade; it stops applying.** For comparison, Yoshida’s positivity theorem reaches $L = \log 2$ and [21] certifies $L = 1.6$.

Ortega-Cerdà and Seip [18] characterize the sampling sequences for PW through de Branges theory — a separated Λ is sampling if and only if it is the zero sequence of $EF + E^*F^*$ for suitable Hermite–Biehler E, F with $H(E) = PW$ — but this is a dichotomy and produces no constant, as the authors state. The one route in that literature which does carry constants, through the A_2 condition on $|G|^2$ for the generating function G , is stated for *complete interpolating sequences*; the zeros are not one, since they overshoot the critical density rather than matching it.

Remark 8.3 (What Suzuki’s de Branges space is, and is not). [16] shows that under RH the Hilbert space of the Weil distribution is isomorphic to a de Branges space — but with structure function $E_\xi(z) = \xi(\frac{1}{2} - iz) + \xi'(\frac{1}{2} - iz)$, whose zero set is not the zeta zeros, and the isomorphism is with the whole Weil space rather than with $PW_{L/2}$. Its chain of subspaces is indexed by the support cut-off, which is structurally our L , and its content is the conversion of Weil’s inequality into an identity. It contains no smallest eigenvalue, no gap and no rate. The two questions are adjacent, not the same, and the de Branges side currently supplies nothing quantitative. Earlier drafts of this program said otherwise.

8.5 Where the obstruction lives, and a construction that fails

Let $\lambda_L^{(W)}$ be the minimum of the Rayleigh quotient over trial functions whose spectral content is confined below W . The fraction of the exponent it recovers is, measured at six lengths:

W/τ_c	0.50	0.75	1.00	1.25	1.50	2.00
$L = 1.6$	0.585	0.733	0.858	0.933	0.982	0.995
$L = 2.0$	0.561	0.725	0.840	0.910	0.957	0.995
$L = 2.2$	0.586	0.744	0.842	0.921	0.969	0.997
$L = 2.5$	0.584	0.739	0.835	0.915	0.965	0.996
$L = 2.8$	0.590	0.744	0.844	0.924	0.966	0.997
$L = 3.0$	0.592	0.741	0.843	0.915	0.964	0.997

(entries $\log \lambda_L^{(W)} / \log m_L$). The columns are constant to about one percent while m_L moves through 10^{80} : **the frequency profile of the obstruction is universal in W/τ_c** , so (8.2) is the right scale, and the obstruction is essentially complete at $2\tau_c$, between τ_c and T_* . There is a practical corollary, and it is the reason the numbers below are reported with a basis-convergence rule rather than as single values: a trial space must reach about $2.5\tau_c$, that is a dimension of about $5Le^L$, or the computed value saturates and looks converged when it is not.

The construction the deficit (8.3) suggests — stay below τ_c and *vanish* at the $2N(\tau_c)$ sub-critical zeros, leaving $D(L)$ free dimensions — does not work, and fails worse as L grows:

L	1.6	2.0	2.4	2.8	3.0
$\log_{10} m_L$	−16.75	−29.17	−48.04	−76.75	−96.25
band $ \tau < \tau_c$ only	−14.30	−24.46	−40.41	−64.84	−81.19
band and vanishing	−14.30	−24.46	−36.49	−45.62	−50.64
fraction recovered	0.85	0.84	0.76	0.59	0.53

The minimizer does not interpolate the undersampled zeros; it makes $\hat{F}$ *small at every zero*, including the oversampled ones above τ_c , and imposing exact vanishing is an expensive constraint. In the language of time–frequency localization this construction is $1 - \lambda_n$ for the prolate operator at Shannon number $\mathcal{N}_L = 2Le^L$ and index $\mathcal{N}_L - D(L)$, hence at occupancy fraction $u_L = 1 - D(L)/\mathcal{N}_L$ bounded away from both endpoints. That regime is neither the classical fixed-index one nor the Landau–Widom plunge. [22] already identifies prolate eigenfunctions as the source of very small eigenvalues of the Weil form, without a rate; the sharpest available statement about the rate is $-\log(1 - \lambda_n) \asymp (c - n)/\log \frac{2c}{c-n}$ with unequated constants, and the corresponding limit at fixed occupancy is an open problem [20].

8.6 The collapse is set by the density, not by the arithmetic

The natural control is a point set with the *same counting function* and no arithmetic. (Deleting the prime comb is not such a control: archimedean plus pole is not a sum of squares over a point set, and is in fact indefinite for $L \gtrsim 0.85$.) We use the Gram points g_n , defined by $\theta(g_n) = n\pi$, whose counting function is exactly the smooth part $\theta(T)/\pi + 1$ with $S \equiv 0$; and jittered Gram points, the jitter applied in the unfolded variable θ/π so that the counting function is preserved. Everything is computed in the same basis, at the same truncation, and at the same precision.

L	$\log_{10} m_L$ (the zeros)	$\log_{10}$, Gram points	ratio of logarithms
0.6	−2.11	−1.00	2.100
0.8	−3.73	−2.17	1.715
1.0	−6.01	−4.06	1.480
1.2	−8.76	−6.62	1.323
1.4	−12.35	−9.97	1.239
1.6	−16.75	−14.18	1.181
1.8	−22.35	−19.49	1.147
2.0	−29.17	−26.15	1.116
2.2	−37.62	−34.48	1.091
2.4	−48.04	−44.77	1.073
2.6	−60.97	−57.48	1.061
2.8	−76.75	−73.17	1.049

At $L = 2$ the four arithmetic-free sets — Gram, and jitter of 0.15, 0.40 and 1.00 of a mean spacing — give $\log_{10}$ of -26.15 , -25.99 , -26.53 and -25.44 against -29.17 for the zeros; at $L = 1.6$, -14.18 , -14.02 , -14.59 and -13.50 against -16.75 . Three readings.

1. **Every arithmetic-free set of the same density collapses superexponentially, at essentially the same rate.** The ratio of logarithms falls monotonically from 2.100 at $L = 0.6$ to 1.049 at $L = 2.8$.
2. **The arithmetic is worth about three orders of magnitude and grows like $\log L$,** against a total growing like e^L . It is a systematic bias and not noise: jitter of up to a full mean spacing moves $\log_{10} m_L$ by about one in either direction, and the zeros sit below the whole arithmetic-free family.
3. **So the margin carries almost no arithmetic information.** Its collapse is what the density $\frac{1}{2\pi} \log \frac{\tau}{2\pi}$ does to the lower sampling constant of a Paley–Wiener space whose type is far below it.

Remark 8.4 (Scope of the measurement). The values for the control sets are trial-space Rayleigh quotients with the sum truncated, hence upper bounds on their own infima and slightly depressed by truncation, so the comparison is conservative in the direction that matters: the control values could only be higher, which would widen the gap without changing its relative decline. The zeros row is computed from the explicit formula without truncation. One jitter draw per strength; the spread across strengths is the error bar quoted, not a statistical estimate. The range is $L \in [0.6, 3.0]$, which is 96 orders of magnitude in m_L but only a factor of five in L , and the ratio of logarithms has not been shown to converge.

8.7 What this changes

Section 7 ended by naming the collapse of the zero-placement precision as the mechanism question. Proposition 7.6 stands and §8.6 does not contradict it; what it does is remove the reason to ask. A Gram-point analogue of ξ , with no primes anywhere in it, has a margin that collapses at the same rate to within a ratio of logarithms tending to one.

Remark 8.5 (Consequence for the program). Two things follow, and they point in opposite directions. Reproducing the superexponential collapse is *weak evidence* for a candidate source, because the density alone produces it; and failing to reproduce it is weak evidence against one. But a candidate is correspondingly *not* required to produce an arithmetic rate. It has to produce a density — a group delay growing like $\log \tau$, which is part of Condition 10.1 — and that is a far cheaper demand than a superexponential margin law. The quantity that does carry arithmetic is the *gap* between the zeros and matched-density sets, about 10^3 at $L = 2$; nothing in this program or in the literature has studied it.

Remark 8.6 (On decay laws, and a convention). No one-parameter law in $D(L)$, in $\mathcal{N}_L$, in $\mathcal{N}_L / \log \mathcal{N}_L$, in $N(\tau_c) / \log N(\tau_c)$ or in $D / \log(2\mathcal{N}_L / D)$ fits $L \in [1.4, 3.0]$ better than 22% in the exponent. Over any factor-1.25 window in L all of them fit to a few percent, which is why fitting on three or four points determines nothing. Two published fits illustrate this: the relation recorded in [13], refitted on its own three horizons, predicts $L = 3.0$ to within 4.1% and $L = 4.0$ to within 6.5%; the Landau–Widom form of [21], fitted on $L \in [2.8, 4.0]$, is 46% high at $L = 1.6$.

A convention warning, because it has already cost this program one wrong conclusion. [21] supports f in $[-L, L]$ and sums primes over $\log n < 2L$; this manuscript supports f in $(-L/2, L/2)$ and sums over $n < e^L$. **The L of [21] is half the L used here.** With the conversion applied,

its certified enclosure $8.9 \times 10^{-18} \leq m \leq 2.27 \times 10^{-17}$ belongs at our $L = 1.6$, where the value computed here is 1.73×10^{-17} , and its 3.19×10^{-283} at our $L = 4$.

Remark 8.7 (Status of the numbers in this section). Propositions 8.1 and 8.2 are written proofs, the second conditional on RH and quoting the Cartwright density theorem. Equations (8.2)–(8.4) are written computations, verified in Appendix A. Every table is a labelled numerical computation, produced by unregistered `mpmath` programmes recorded in [23]; each entry is a trial-space Rayleigh quotient and hence a one-sided bound, and the direction is stated where a comparison is made. The assembly those programmes use agrees with an independent quadrature assembler of this program to a relative 6×10^{-25} , and reproduces the three margins of [13] to 2.2%, 3.2% and 0.2% — which, incidentally, puts Proposition 7.6 on firmer ground than the three data points it was stated with. Section 8.4 is a reading of the cited literature.

9 The endpoint direction

Theorem 5.2 sends us transverse to ω , and the proposal’s own phrase — the variation of the endpoints — names the direction. The compression P_L is not in the commutative algebra: the projections commute with one another, but not with the transfers. Proposition 4.5 has already identified the leading endpoint as the point through which the contraction defect’s flux passes; this section finds the same point from the geometry, works out the algebra there, and reaches the object the investigation is left with.

9.1 The variation is rank one, at the leading endpoint

Let V_ω denote the uncompressed causal transfer, so that $V_{\omega,L}f = P_L V_\omega f$ for f supported in I_L . Fix $f \in C_c^\infty(I_{L_0})$ and let $L > L_0$, so that f is held fixed while the interval grows.

Proposition 9.1 (Endpoint variation). *For $L > L_0$ and $f \in C_c^\infty(I_{L_0})$,*

$$\frac{\partial}{\partial L}(V_{\omega,L}f) = \frac{1}{2} (V_\omega f)(L/2) \delta_{L/2}, \quad (9.1)$$

a rank-one variation supported at the single point $x = L/2$, whose coefficient is the boundary value of the evolved field. Equivalently $\partial_L V_{\omega,L} = B_L V_{\omega,L}$ with $B_L = \frac{1}{2} \delta_{L/2} \otimes \text{ev}_{L/2}$.

Proof. $V_{\omega,L}f = \mathbf{1}_{I_L} \cdot (V_\omega f)$ and $\partial_L \mathbf{1}_{(-L/2, L/2)} = \frac{1}{2}(\delta_{L/2} + \delta_{-L/2})$, so the variation has the two endpoint terms. Causality kills the second: f is supported in I_{L_0} and the kernel of V_ω is supported in $[0, \infty)$, so $\text{supp } V_\omega f \subseteq [-L_0/2, \infty)$ and $(V_\omega f)(-L/2) = 0$ for $L > L_0$. The surviving coefficient is the boundary value of $V_{\omega,L}f$ from inside, since $V_\omega f$ and $V_{\omega,L}f$ agree on I_L . $\square$

Equation (9.1) is exactly the endpoint equation of a line: the variation is local, it sits at the moving end, and its coefficient is the transported field there. The tail is fixed and the head moves, which is what causality means here. This is the closest structural match the proposal has, and it is not an analogy: the two statements have the same form.

Remark 9.2 (The qualification, and it matters). Proposition 9.1 is a *partial* derivative, at fixed ω ; the reason the endpoint term is the whole of it is that V_ω carries no L , so all the L -dependence sits in the indicator. Along a path $t \mapsto (\omega(t), L(t))$ the chain rule gives

$$\frac{d}{dt} V_{\omega,L} = (\dot{L} B_L - \dot{\omega} G_{\omega,L}) V_{\omega,L},$$

whose second term is the full causal generator — non-local, carrying the prime atoms. **Along a path the motion is endpoint *plus* bulk**, and the picture of a dragged endpoint describes one direction of the plane only. What survives without qualification is the stronger, non-differential statement (4.3): the defect *itself* is a boundary flux at every (ω, L) .

That is also the two-term structure a deformed Wilson line has in the first place, an endpoint term and a shape term. Theorem 5.2 removed the shape term in the ω -direction alone; the mixed motion puts a second term back, and Section 9.2 shows it does not commute with the first.

9.2 The algebra at the endpoint, and flatness

It is tempting to expect a curvature in the (ω, L) plane and to look for it in the mixed derivative. That expectation is wrong, and the correction is what sends the search somewhere else. The algebra is small enough to write down.

Proposition 9.3 (The endpoint algebra). *With $B_L = \frac{1}{2} \delta_{L/2} \otimes \text{ev}_{L/2}$:*

1. $\partial_\omega B_L = 0$;
2. $G_{\omega,L} B_L = 0$, *exactly*;
3. *hence* $[G_{\omega,L}, B_L] = -B_L G_{\omega,L}$, *which does not vanish*;
4. *and* $\partial_L G_{\omega,L} = B_L G_{\omega,L}$.

Proof. (i) B_L is built from the geometry of the interval and carries no shift. (ii) $G_{\omega,L} B_L f$ is a multiple of $P_L(g_\omega * \delta_{L/2})$, and $(g_\omega * \delta_{L/2})(x) = g_\omega(x - L/2)$ vanishes for $x < L/2$ because g_ω is causal, hence vanishes on I_L up to a null set: the endpoint operator injects at the leading edge and a causal generator transports only to the right of it, which is outside the interval. (iii) is immediate from (ii), and $B_L G_{\omega,L} f = \frac{1}{2}(g_\omega * f)(L/2)\delta_{L/2}$ is not zero. (iv) $G_{\omega,L} f = \mathbf{1}_{I_L} \cdot (g_\omega * f)$ for f supported in I_{L_0} and $L > L_0$, so differentiating the indicator as in Proposition 9.1 gives $\partial_L(G_{\omega,L} f) = \frac{1}{2}(g_\omega * f)(L/2)\delta_{L/2}$, which is $B_L G_{\omega,L} f$. $\square$

The commutator is *one-sided*: the pair generates a triangular rather than a commutative algebra. So the non-commutativity that Remark 5.5 demanded transverse to ω does exist. What it does not do is produce a curvature.

Proposition 9.4 (Flatness). *On the range of the two-parameter family, wherever $V_{\omega,L} f$ is twice continuously differentiable in (ω, L) ,*

$$\partial_\omega B_L + \partial_L G_{\omega,L} + [G_{\omega,L}, B_L] = 0. \quad (9.2)$$

Proof. Mixed partial derivatives of $V_{\omega,L} f$ commute; substituting $\partial_\omega V_{\omega,L} = -G_{\omega,L} V_{\omega,L}$ from (2.15) and $\partial_L V_{\omega,L} = B_L V_{\omega,L}$ from Proposition 9.1 and collecting terms gives (9.2) applied to $V_{\omega,L} f$. Alternatively, and this is the content of the identity, substitute Proposition 9.3: (i) kills the first term and (iii) turns the third into $-B_L G_{\omega,L}$, so (9.2) is (iv). $\square$

So the pair $(-G_{\omega,L} d\omega, B_L dL)$ is a non-abelian but **flat** connection wherever it is a connection at all, and the flatness is not an extra hypothesis but the explicit identity $\partial_L G_{\omega,L} = B_L G_{\omega,L}$: the generator's own L -variation is its boundary value. There is no local curvature to compute and no surface term to extract, and on a simply connected parameter domain the holonomy of a flat connection depends on the endpoints alone. Appendix A verifies (ii), (iii) and the integrated form of (iv) as exact finite algebra.

9.3 The connection does not close, and it misses by exactly the borderline

Proposition 9.1 holds on the core $C_c^\infty(I_L)$, where $V_\omega f$ is smooth because $\hat{f}$ decays rapidly. It does not extend to the form domain.

Proposition 9.5 (No form-bounded endpoint evaluation). *B_L is not bounded relative to $Q_{0,L}$, at any $0 \leq \omega \leq \frac{1}{2}$.*

Proof sketch. B_L is built from $\text{ev}_{L/2}$, and a point evaluation on $\mathbb{R}$ is bounded on H^r only for $r > \frac{1}{2}$. The form domain is $\mathcal{D}_{\log,L}$ of Definition 2.1, whose multiplier $b(\tau^2) = \log |\tau| + O(1)$ is only logarithmically stronger than L^2 ; it is contained in no H^r with $r > 0$ and carries no trace operator, which is also why the natural closed domain of the archimedean energy has no trace condition [2]. The transfer does not repair this: the bound $|K_\omega(\eta + i\tau)| \leq C_\eta(2 + |\tau|)^{-\omega}$ of [1] gives V_ω a smoothing gain into H^r for $0 < r < \min(\omega, \frac{1}{2})$ only, which is strictly below the trace threshold for every admissible shift, the endpoint $\omega = \frac{1}{2}$ included. $\square$

Remark 9.6 (A third marginality at the same exponent). The failure in Proposition 9.5 is not by a wide margin; it is exactly borderline. Three separate quantities in this program sit at the same exponent and on the wrong side of it: the archimedean kernel $n_\gamma(r) \sim \frac{1}{2r}$ makes K the Gagliardo seminorm at $s = 0$, the borderline of the scale, so the form domain is only logarithmically better than L^2 ; the transfer gains at most $(2 + |\tau|)^{-1/2}$; and a trace needs $r > \frac{1}{2}$. We record the coincidence and do not explain it. Whether it is one fact seen three times is Problem 9.8.

9.4 The primes are the atoms of the connection

The proof of Theorem 3.1 decomposes the kernel of $G_{\omega,L}$ into a continuous part and an atomic part:

$$\begin{aligned} g_\omega(t) &= g_\omega^{\text{cont}}(t) + g_\omega^{\text{at}}(t), \quad t > 0, \\ g_\omega^{\text{cont}}(t) &= 4 \cosh(\omega t) \cosh(t/2) - 2 \cosh(\omega t) n_\gamma(t) + \text{const}, \\ g_\omega^{\text{at}}(t) &= -2 \sum_{n \geq 2} \Lambda(n) n^{-1/2} \cosh(\omega \log n) \delta_{\log n}, \end{aligned} \tag{9.3}$$

the atoms sitting at the prime periods $m \log p$ with masses $2(\log p)p^{-m/2} \cosh(\omega m \log p)$. This is the Lévy decomposition that the companion investigation's ninth selection rule requires of any source [3], and it is here a property of the connection rather than of a prepared state.

Read along the line, an atomic connection is a connection with *punctures*: the holonomy of the segment is an ordered product of one local factor per atom crossed, times the contribution of the continuous part. That is precisely the periodic-orbit structure the program's own analysis says the primes demand and which a short-distance mechanism cannot supply [5]. It is also, given Propositions 9.4 and 9.5, the only place left for the proposal's mechanism to live: a flat connection has no local curvature, but a flat connection on a punctured domain has monodromy, and monodromy is exactly the non-local, potentially non-abelian datum the proposal needs.

The obstruction is stated by Theorem 5.2. The local factors at different atoms commute, because every object here is a convolution and the atomic masses are *scalars*. The monodromy representation is therefore one-dimensional and carries no more information than the product of its masses. Proposition 9.3 sharpens what is needed: the algebra generated by the bulk and endpoint directions *is* non-abelian, so the missing ingredient is not non-commutativity but a reason for the holonomy to see it — and by Proposition 9.4 there are exactly two, namely non-simple-connectivity and failure of the connection to exist. To make it carry more, the residues at the atoms would have to be matrix valued — and the arithmetic object whose prime sum carries matrix-valued residues

is $\text{tr } \rho(\text{Frob}_p)$ for a representation ρ . The three strands of Remark 5.5 converge on this point: the endpoint direction is where the geometry is, the atoms are where the primes are, and an internal index at the atoms is the only way to make the two non-abelian at once. We state this as the structural conclusion of the paper and not as a programme, since it changes the target from ζ to a family.

9.5 Two open problems, and why they are one

Remark 9.7 (Flatness concerns the slice, not the space of contours). Proposition 9.4 is about a *two-parameter* family. The space of contours is infinite dimensional and the conformal group acts on it with finitely many parameters, so flatness of a two-dimensional slice says nothing about the rest. Put usefully: if a correspondence between contour deformations and motions in (ω, L) exists it is many-to-one, and the fibre — contour deformations that do not move (ω, L) — is untouched here. That fibre is exactly where the shape term of the non-Abelian Stokes variation acts, a deformation at fixed endpoints being an area deformation. So Theorem 5.2 constrains motion along the slice and not the shape term, which acts on the fibre. Whether it acts trivially there is uncomputed: trivially, and the correspondence is consistent; nontrivially, and it is overdetermined, which is a condition on a candidate rather than a freedom.

Problem 9.8 (Close the endpoint flow, or show it cannot be closed). Proposition 9.5 obstructs B_L on the form domain while Proposition 9.1 defines it on a dense core. Determine whether the endpoint flow admits a closed realization on any domain on which $Q_{\omega,L}$ is also closed. Section 10 recasts this as a smearing question, with the profile fixed by Condition 10.3: an endpoint operator in a field theory is a distribution and must be smeared, and the required effective dimension $\Delta = \frac{1}{2}$ is the tightest smearing the domain permits.

Problem 9.9 (Monodromy of the punctured flow). Compute the local factor of (9.3) at a single atom, as an operator on the core, and determine what a matrix-valued residue at that atom would have to satisfy for the resulting holonomy to remain a contraction in the sense of Proposition 2.2.

These are not independent. Proposition 9.3 says the two-parameter connection is non-abelian; Proposition 9.4 says it is flat; and a flat non-abelian connection on a simply connected domain where it exists has trivial holonomy. So there are exactly two ways for the structure found in Proposition 9.3 to become visible, and they are precisely the two problems: **non-simple-connectivity**, which is the punctures of Section 9.4 and is Problem 9.9; and **failure of the connection to exist**, which is Proposition 9.5 and is Problem 9.8. That is a tighter position than the investigation had before: the mechanism is no longer missing, it is present and trivialized, and there are two named ways to untrivialize it.

Remark 9.10 (What the L -direction is not). Two cautions, both of which this program has paid for once. First, Q_L is not a family of functionals but one functional restricted by support, so a realization at fixed L carries no information and an L -family of unrelated realizations proves nothing [2]. The L -dependence that is real is in the compression, which is what Proposition 9.1 differentiates. Second, the exclusions this program has accumulated are statements about channel decompositions; a flow is not a decomposition, so they neither block this direction nor may be quoted in its support [4].

10 Conditions on a mechanism

A candidate arising from this picture inherits the necessary conditions the companion investigation has accumulated [3]. We do not restate their derivations; the table records which bear on a flow,

and what this paper has added to each.

Inherited condition	Bearing, after this paper
Reflection invariance	The flow must commute with $x \mapsto -x$; the involution is the functional equation, which is also what makes Proposition 4.1 true.
Unboundedness relative to L^2	Now a statement about <i>time</i> : by Proposition 4.3 the delay grows like $\log \tau$, so a packet at frequency N is held for $\sim \log N$. Condition 10.1 restates the exclusion in the form a candidate can fail.
Spectral weight and channel tower	The archimedean multiplier <i>is</i> the group delay (Proposition 4.3), so the density identity is no longer an input to be matched but a property of a phase.
Criticality	Section 7: the flow formulation inherits it. A bound at fixed ω buys a zero-free strip of width ω and no more.
Jump structure	Realized on the connection by (9.3): the atoms of the kernel are the prime periods.

Three conditions are specific to the deformation picture. They are *proposals*: each is elementary, none has been tested against a specific theory, and the third carries an identification hypothesis this program deliberately does not make.

10.1 The object is a bilinear, not a character

Condition 10.1 (Bilinear endpoints, and a delay test). The object must be a bilinear in unbounded endpoint operators joined by a transport — not a character — and its group delay must grow like $\log \tau$ with smooth part $2\theta'(\tau)$.

The first clause names a class and the second is the test. The exclusion behind the clause is worth restating precisely, because it is narrower than it looks. A holonomy valued in a compact group, in a finite-dimensional representation, is unitary, so its character is a bounded function and every pairing assembled from it is bounded on $L^2(I_L)$ — excluded by the unboundedness rule. That argument uses only that the object is a *trace of a group element*. It is also the structural reason the conformal Wilson benchmark of [3] failed: $2 \cos \theta$ is bounded *because* it is a character of $SU(2)$ in the fundamental, not by accident of that benchmark.

An *open* Wilson line is not a character. In the construction of [15] the operator is

$$OW_i^j[\tilde{C}] = \bar{\psi}^j(x_j) P \exp \left[\int_{\tilde{C}} ds \left(i A_\mu \dot{x}^\mu + n^k(s) \Phi_k |\dot{x}| \right) \right] \psi_i(x_i),$$

a bilinear in matter fields joined by a Wilson line, the endpoint fields transforming in the fundamental and antifundamental so that the contraction is gauge invariant. Its endpoints carry operator-valued distributions, so the object is unbounded by construction and the character argument has nothing to act on. **The condition is satisfied structurally rather than evaded**, which is why it is stated here as a prescription rather than a prohibition.

Two further remarks connect that construction to Section 9. First, the geometry matches: the supersymmetric configuration of [15] in the defect theory is a *ray ending on a defect*, one end pinned and one end free, and Proposition 9.1 found the variation to be rank one at the leading end only, the trailing one killed by causality — with the boundary value of the evolved field as the coefficient,

which in the open Wilson line is the endpoint field itself. Second, the obstruction of Proposition 9.5 is, in that language, the ordinary fact that a field at a point is a distribution and must be smeared; Condition 10.3 fixes the profile.

Note also that $V_{\omega,L}$ is not unitary — it is a contraction and the object of interest is its defect (2.16) — so the transport to be matched is not anti-Hermitian either.

10.2 The non-commutativity must be made visible

Condition 10.2 (Visible transverse structure). A mechanism must supply a reason for the holonomy to see the non-commutativity of Proposition 9.3: either a domain that is not simply connected, or a failure of the connection to exist.

This replaces the weaker demand of Remark 5.5, which asked only for non-commutativity transverse to ω . That exists (Proposition 9.3); what it lacks is visibility (Proposition 9.4). Problems 9.8 and 9.9 are the two forms the condition can take.

10.3 Marginal short distance: effective dimension one half

Condition 10.3 (Marginality). Suppose the line's parameter is identified with the arithmetic coordinate x . Since $n_\gamma(r) = \frac{1}{2r} + O(1)$ as $r \downarrow 0$ by (2.2), (2.3) gives

$$K[F] = \frac{1}{2} \iint |F(x) - F(y)|^2 n_\gamma(|x - y|) dx dy \sim \frac{1}{4} \iint \frac{|F(x) - F(y)|^2}{|x - y|} dx dy$$

near the diagonal: the Gagliardo seminorm at the borderline exponent $s = 0$. An insertion on the line reproducing this must have two-point function $|x - y|^{-1}$, an effective defect dimension $\Delta = \frac{1}{2}$, with the coefficient fixed.

This excludes the first thing one would try. The displacement operator of a line defect has protected dimension $\Delta = 2$, fixed by the Ward identity that defines it, and the standard protected scalar insertions on a supersymmetric Wilson line have $\Delta = 1$ [12, 11]. *So the deformation whose generator matches is not the displacement operator*, which is what a deformation of the contour would naively produce.

Remark 10.4 (The principal series, and why everything here is marginal). There is one reading under which Condition 10.3 is not an accident. For a one-dimensional conformal system the dilatation eigenvalue is the conformal weight, and the unitary principal series of $SL(2, \mathbb{R})$ is $\Delta = \frac{1}{2} + i\tau$ with τ real — the critical line. In that reading the zeros are principal-series data, the shift ω moves off the unitary line, which is the detection threshold of Corollary 7.3, and the Riemann hypothesis is the statement that nothing leaves it. Condition 10.3's $\Delta = \frac{1}{2}$ is then the *edge* of the principal series, the most marginal unitary weight there is, which would explain the coincidence of Remark 9.6: the Gagliardo borderline, the transfer's $(2 + |\tau|)^{-1/2}$ gain and the trace threshold $r > \frac{1}{2}$ are three faces of one exponent. Section 10.4 supplies the missing motivation: on a ray, the Mellin variable is the dilation eigenvalue and $s = \Delta + i\tau$, so $\Delta = \frac{1}{2}$ puts it on the critical line — and $\frac{1}{2}$ is what Condition 10.3 requires for an unrelated reason. The identification of the arithmetic coordinate with $\log r$ is therefore *derived from a symmetry match* rather than assumed: if one asks for a geometry whose residual conformal symmetry is the Weil form's exact symmetry group, the ray with a defect at one end is that geometry. *The program's caveat is not removed — this remains an identification — but there is now a reason to make it. The neighbourhood, conformal quantum mechanics and the zeros, is crowded.*

10.4 Conformal symmetry: the form already has it

It is natural to ask whether the conformal symmetry of Yang–Mills could be used to compactify and tame the infinities of the problem. The answer is better than the question expects, and it corrects a claim made in version 0.2 of this manuscript.

The form’s exact symmetry group is a conformal one. Every term of (2.6) is either a multiple of $\|f\|^2$ or an integral against a kernel depending on $x - y$, so the Weil functional is invariant under $f \mapsto f(\cdot - a)$, exactly and unconditionally — the Toeplitz structure the companion investigation’s numerics exploit. Write $x = \log r$, so that the arithmetic coordinate is the logarithm of a radial coordinate on a ray.

in x	in r	status for the Weil form
translation $x \mapsto x + a$	dilation $r \mapsto \lambda r$	exact symmetry
reflection $x \mapsto -x$	inversion $r \mapsto 1/r$	exact symmetry (rule 1)
dilation $x \mapsto \lambda x$	$r \mapsto r^\lambda$	not a symmetry, and not conformal

And L is a modulus, not a length. In the ray coordinate the interval I_L is the annulus $e^{-L/2} < r < e^{L/2}$, of ratio e^L and conformal modulus $L/2\pi$. Dilations move it and preserve L , and nothing else about it is conformally meaningful, so the program’s second parameter is exactly the one conformal invariant of the configuration. This is the consistency check the reading below needs: had L been a scale, the dilation invariance of the form would have been in conflict with the whole L -dependence.

The Möbius maps of the line preserving $\{0, \infty\}$ are exactly $r \mapsto \lambda r$ and $r \mapsto \mu/r$, so the stabilizer of a ray’s two endpoints is generated by dilation and inversion — and the Weil form is invariant under all of it. The special conformal generators move the endpoints and are broken by the configuration, not by the form. Hence:

the Weil form’s exact symmetry group is precisely the residual conformal group of a ray with one end pinned at the origin and the other at infinity,

which is the configuration of [15]. Both halves of that symmetry were already program facts; what is new is that they are the stabilizer of a geometry.

Correction. Version 0.2 argued here that exact scale invariance is incompatible with the target, citing the exclusion of [4] that dilation destroys the prime atoms. That exclusion concerns dilation *in x* , which is $r \mapsto r^\lambda$ and is not a Möbius map. Under the conformal dilation $r \mapsto \lambda r$ the atoms sit at fixed *ratios* $r_2/r_1 = p^m$ and are preserved. In the ray picture the Weil form has no absolute scale to protect, and conformal invariance is not in conflict with it. What the atoms require is a preferred set of ratios, which is a statement about the spectrum of dimensions rather than about scale: in Mellin variables the prime part carries weight $-\zeta'/\zeta$, a continuum, and a conformal system with a continuous spectrum of dimensions is exactly what the principal series is. The demand is therefore not *break conformal invariance* but *carry this continuum of dimensions*, which is sharper and more useful. We do not challenge the exclusion of [4] in its own setting, where the relevant dilation is the field theory’s own.

Positivity is a conformal-orbit invariant.

Proposition 10.5 (Congruence). *If M is invertible on the form domain then $Q \succeq 0$ if and only if $M^*QM \succeq 0$. A conformal reparametrization acts on a weight- Δ kernel by congruence with multiplication by a positive power of the Jacobian, so positivity of the target is invariant along its conformal orbit.*

This is Sylvester’s law of inertia. Since the criterion asks only for a sign, any member of the conformal orbit of Q_L is an equally good target, and the ray-to-semicircle map of [15] is not merely a reframing: the semicircle form is a genuinely different quadratic form with the same positivity content, on a *compact* contour. The program already uses one congruence of this kind, the compression presentation $Q_L = K_+ - T_L$ read as $T_L \preceq K_+$; conformal congruences are a second, geometrically natural class that has not been used.

Two guards. The freedom is vacuous if the class is unrestricted, since every positive form is congruent to the identity — any use must name the class in advance. And the Jacobian of the ray-to-semicircle map *degenerates at the endpoints*, so the congruence is boundedly invertible in the interior and fails exactly where Proposition 9.5 already obstructs. Whether that degeneration softens the endpoint or leaves the form domain is open, and it is the first thing to check if this route is pursued.

What compactifies. The spectral infinity does, and the compactification is already in hand: the conformal map of the right half-plane to the disc is the Cayley transform of Proposition 7.8, under which K_ω becomes an inner function of the disc whose zeros accumulate at one boundary point, the logarithmically divergent delay of Proposition 4.3 being exactly that accumulation. The length infinity does too, by the ray-to-semicircle map. What does not is Proposition 9.5, a short-distance statement invariant under reparametrization; and Corollary 7.3, a statement about the zero set of ξ , so compactifying moves its content to a boundary point without removing it.

The vanishing theorem, read the other way. [15] finds that in non-superconformal $\mathcal{N} = 2$ theories with fundamental hypermultiplets all BPS open Wilson lines have vanishing expectation value, nontrivial protected operators appearing only in the superconformal case. Version 0.2 read this as a trap. It is not: superconformality is available and is not in conflict with the target, because the target’s exact symmetry group is the residual conformal group of the geometry in which those operators live. That makes the superconformal defect setting the natural place to run the test of Condition 10.1, rather than a place where it was expected to fail for structural reasons.

Where the ratios could come from. Not from a single conformal defect, which preserves dilations about a point on it and therefore fixes no ratio. Ratios are *separations*: the atoms sit at $r_2/r_1 = p^m$, so what is wanted is a spectrum of dimensions carrying weight $-\zeta'/\zeta$, equivalently a structure invariant under $r \mapsto \lambda r$ but distinguishing the ratios p^m . That is (9.3) in geometric dress, with the punctures at integer ratios, and it is the same statement as Condition 10.2. The named object in that direction is the Bost–Connes system [27]; we record where the structure lives and do not enter that literature.

11 Outlook

11.1 What is settled

Three things are now closed, and it is worth separating them from what is not.

The generator and the transfer are identified. Theorem 3.1 writes the form in closed shape and shows the whole shift dependence is one hyperbolic factor; Proposition 4.1 shows the transfer is an all-pass filter, exactly and unconditionally. Everything in Sections 4–7 follows from the second.

Two of the program's own facts are explained rather than recorded. The density identity $b(\tau^2) + w_0 = 2\theta'(\tau)$ is the statement that the Weil symbol is a group delay (Proposition 4.3), and the band edge at γ_1 of the seventh selection rule is the statement that the cheapest way to avoid every zero is to live below the lowest one (Section 7).

The shift–length plane is settled. Corollary 7.3 makes the shifted family a graded criterion, one zero-free strip per shift; the contraction region has no boundary under the Riemann hypothesis, so the critical path sought by [13] does not exist; and Remark 7.5 says what the plane is for instead.

The margin is identified, is positive, and is not arithmetic. It is the lower sampling constant of the zeros on $PW_{L/2}$, with no upper one (Proposition 8.1); it is strictly positive because the group delay diverges (Proposition 8.2); and its superexponential collapse is reproduced, to within a ratio of logarithms tending to one, by an arithmetic-free point set of the same counting function (§8.6). That closes the question Section 7 left open, in the sense of answering it rather than of resolving it in the direction expected.

11.2 What is open, ranked

1. **The group-delay test on a specific open Wilson line.** Condition 10.1 is one computation: take a protected open Wilson line two-point function in the defect setting of [15], extract the phase of its spectral kernel against the dilatation eigenvalue, and compare the derivative with $2\theta'(\tau)$. A bounded delay kills the candidate; a delay growing like $\log \tau$ with the wrong coefficient kills it too. This needs no arithmetic and no positivity, and it is the first item because it is the first test in this program that a gauge theory can fail.
2. **What does the arithmetic control, if not the margin?** §8.6 measures the gap between the margin of the true zero set and the margins of matched-density arithmetic-free sets: about three orders of magnitude at $L = 2$, growing like $\log L$ against a total growing like e^L . That gap is a well-defined and computable quantity which *does* carry arithmetic, and neither this program nor the literature has studied it. It replaces the previous second item, which asked how ε_L collapses; that question now has an answer with no arithmetic in it.
3. **The universal frequency profile of §8.5.** The obstruction's spectral content is a fixed profile in τ/τ_c , stable to about one percent while the margin moves through 10^{80} . That is the one structural regularity the measurement turned up that is not explained here, and an exact statement of it would be worth having.
4. **Problems 9.8 and 9.9**, which Section 9 shows are the two forms of one question.
5. **Does the vanishing theorem generalize?** Whether "BPS implies vanishing expectation unless superconformal" extends beyond the class of [15]. If it does, it is a general obstruction of the same kind as this program's own exclusions and belongs in the selection index; if it does not, the exception is where to look.
6. Only after these, a specific theory. Section 10 says what it must satisfy, and Remark 6.3 says where to look first.

11.3 Scope

Nothing above is proposed as a route to the Riemann hypothesis, and Section 7 explains structurally why no argument of this shape could be one.

Written proofs: Proposition 8.1, Proposition 8.2, Theorem 3.1, Proposition 4.1, Proposition 4.3, Proposition 4.4, Proposition 4.5, Lemma 5.1, Theorem 5.2, Proposition 6.1, Proposition 7.1, Proposition 7.2, Corollary 7.3, Proposition 7.8, Proposition 9.1, Proposition 9.3 and Proposition 9.4.

Conditional on the Riemann hypothesis: Proposition 4.5, Proposition 8.2, the reading (8.1) of the margin as a sampling constant, and the second case of Proposition 7.2, which uses the weaker hypothesis stated there. The computations of Section 8 are themselves unconditional: they evaluate the Weil functional, which is defined without the hypothesis. Proposition 4.4’s passage from one zero to the sum is stated rather than proved uniformly and is supported by Appendix A.

Proof sketch: Proposition 9.5, resting on a smoothing estimate quoted from [1] and on the Sobolev trace threshold.

Labelled numerical computation: every table of Section 8, each entry a trial-space Rayleigh quotient and hence a one-sided bound, produced by unregistered programmes recorded in the investigation’s notes; and the convergence rule of §8.5, which is itself a measurement.

Heuristic, and labelled where it occurs: the single-parameter relation $\kappa_L \sqrt{m_L} \approx \sqrt{Q[Xe_L]}$ of Proposition 7.6, whose support is three data points — independently reproduced to a few percent, per Remark 8.7, but still three; the detection scale $\omega \sim 1/L$ of Remark 7.7; and Remark 10.4, which is a reading resting on an identification the program declines to make.

Not claimed anywhere: a source, a positivity statement, a matched gauge theory, an excluded gauge theory, or novelty for the idea that the zeros behave like resonances.

Preparation

This working manuscript was produced with a large language model, Anthropic’s Claude (model `claude-opus-5`), working directly in the investigation repository: the derivations and proofs of Sections 3–9, the conditions of Section 10, the identification and measurement of Section 8, the two check programs of Appendix A, and the drafting throughout. The proposal the paper analyses is Edward Baker’s, as is the open Wilson line construction of [15] and the earlier shifted-zeta material of [13, 14] on which Section 7 builds; Remark 6.2 records the priority of the hyperbolic identity. The research notes accompanying the investigation name the model used for each and record the exploratory computations deliberately outside the registered checks. Every statement is classified in the text as a written proof, a proof sketch, a labelled numerical computation, a heuristic, or a reading.

A Checks

Two programs,

```

    numerics/check_causal_commutation.py,
    numerics/records/causal-commutation-checks.json,
    numerics/check_sampling_forms.py,    numerics/records/sampling-forms-checks.json,
```

the second of each pair being its preserved record. Both use the standard library only and print JSON to standard output. Exact rational arithmetic is labelled separately from floating-point work. Counts refer to finite test cases, not to independent theorems, and none of the checks is a positivity certificate.

Group	What is verified
Multiplicativity of truncation	Lemma 5.1 as a finite matrix identity, $P_L A P_L B P_L = P_L A B P_L$, in exact rational arithmetic, for randomly generated causal kernels at several truncation sizes.
Commutation, and its cause	$[A, B] = 0$ for truncated causal convolutions, exact (Theorem 5.2); and the control that truncated <i>non</i> -causal convolutions have commutators bounded away from zero, so causality is what the lemma uses.
The causal pole kernel	That $4 \cosh(\omega x) \cosh(x/2) \mathbf{1}_{x>0}$, symmetrized by (3.3), equals $2 \cosh(\omega(x-y)) \cosh(\frac{x-y}{2})$, and at $\omega = 0$ equals P_L in the form $2 C(f) ^2 - 2 S(f) ^2$ (Theorem 3.1).
Vanishing on the imaginary axis	That $\operatorname{Re}[2(p - \frac{1}{2})^{-1} + 2(p + \frac{1}{2})^{-1}] = 0$ at $p = i\tau$, exactly, over a grid (Remark 3.3).
The hyperbolic identity	Proposition 6.1 as a finite matrix identity at several ω and L .
The endpoint algebra	Proposition 9.3 in exact rational arithmetic: $G_{\omega,L} B_L = 0$; the one-sided commutator $[G_{\omega,L}, B_L] = -B_L G_{\omega,L}$ with $B_L G_{\omega,L} \neq 0$; and the integrated form of $\partial_L G_{\omega,L} = B_L G_{\omega,L}$ at two extensions.
The resonance model	Propositions 4.1 and 4.4 in a Blaschke model: that a factor with a zero at $\omega + i\gamma$ is unimodular on the critical line and that $ b-1 ^2$ equals the Lorentzian $4\omega^2/(\omega^2 + (\tau - \gamma)^2)$ <i>exactly</i> , both in exact rational arithmetic; and, in floating point, that its mass is $4\pi\omega$ and that the summed group delay over a window is 2π times the number of resonances in it.

The second program, 336 cases in five groups, supports Section 8.

Group	What is verified
Basis algebra	The closed forms for the autocorrelations S_{jk} and the pole pairings $E_j^\pm$ of the orthonormal sine basis of $L^2(I_L)$, against direct quadrature, at four lengths; and the parity rule.
Archimedean integrals	The three families of integrals against n_γ needed to assemble $Q_{0,L}$, summed in closed form by (2.2) into ψ and ψ' at $\frac{1}{4}$ plus a geometrically convergent remainder, against quadrature; together with $\int_L^\infty e^{-r/2} n_\gamma(r) dr = \operatorname{artanh}(e^{-L})$.
Assembly against the zeros	The assembled $Q_{0,L}$ matrix elements against a direct sum over one hundred published zero ordinates plus a smooth tail, at three lengths. Agreement is 1.8×10^{-4} , limited by the tail, so this checks the assembly and not the arithmetic of ζ .
The sampling budget	(8.3), that $D(L) = 2e^L - \frac{7}{4}$ exactly, against the Riemann–von Mangoldt main term at six lengths; and the count crossover $T_* = 2\pi e^{L+1}$.
Conventions	That the prime mass of [21] at its L is the comb of (2.6) at $2L$, exactly, at three lengths; this is the factor of two of Remark 8.6.

Run from the investigation directory:

```
python3 numerics/check_causal_commutation.py
python3 numerics/check_sampling_forms.py
python3 validation/drafts.py check --replay
```

Scope. The resonance group uses the first twelve zero ordinates, truncated to exact rationals, as *published input data*; the Blaschke identities hold for any rational γ , so the group verifies the mechanism of Section 4 and not the arithmetic. Nothing here computes with ξ or ζ . The checks do not verify the convergence arguments in the proof of Theorem 3.1, which are analytic, nor Proposition 9.5, which is a statement about domains, nor Propositions 7.1–7.2, which are statements about a limit. The numbers quoted in Section 7 from [13] are reproduced, not re-derived, and carry the status recorded where they were first stated. Nor do they verify anything in Section 8 beyond its closed forms and its budget: every table there comes from exploratory computations requiring `mpmath`, which the repository’s conventions exclude, kept outside the registered programmes in `numerics/exploratory/` and labelled there. Those computations agree with an independently written quadrature assembler of this program to a relative 6×10^{-25} , which is a consistency check and not a certificate.

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